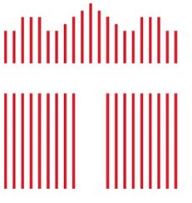
Text as Data

W8: Topic Models and Their Variants

Yen-Chieh Liao

2026-10-30

Fall 2026 (115-1) · National Taiwan University (GIND)
NTLDEV 7191 · Fri. 16:30–18:20 · GIND 204
Last compiled: 2026-11-06 11:16:45 (Asia/Taipei)



National Taiwan University
國立臺灣大學

Political positions and issue attention

Research task	Political question
W7: text scaling	Where does a speaker stand on a defined dimension?
W8: issue attention	What does a political actor choose to discuss?

Campaign promises, newspaper frames, policy diffusion and social media show different uses of topic models.

We begin with each study's political puzzle.

Reference: W7 scaling; Grimmer, Roberts & Stewart (2022)

Research questions and model use

Guiding question: How do political scientists turn large text collections into evidence about political behavior and communication?

We first read seven political studies. Then we unpack how the models work and apply them to political speeches in R.

Reference: Grimmer, Roberts & Stewart (2022); ESS course

Topic models in political research

Case	Texts	How the model is used
1 · Catalinac (2016)	Japanese manifestos	Measure campaign promises
2 · Gilardi et al. (2021)	U.S. news paragraphs	Study policy diffusion and framing
3 · Barberá et al. (2019)	Tweets	Measure daily issue attention
4 · Jiang et al. (2024)	Hong Kong news	Develop and test a hypothesis
5 · Roberts et al. (2014)	Survey answers	Compare priorities with human coding
6 · Mueller & Rauh (2018)	Newspaper articles	Predict next-year conflict

Reference: Catalinac (2016); Gilardi, Shipan & Wüest (2021); Barberá et al. (2019); Jiang, Xia & Zhang (2024); Roberts et al. (2014); Mueller & Rauh (2018)

How we will read each case

What to find	Question for the reader
Political question	What puzzle does the study want to explain?
Texts and document unit	What counts as one document, and whose language appears?
Model contribution	What does the topic model discover or measure?
Evidence and limits	What comparison, dynamic model or experiment supports the claim?

Use the same four questions when you design your own text research.

Teaching guide for the six cases introduced on the previous page.

Electoral rules and campaign promises

Case 1 · Campaign priorities



Amy Catalinac

Journal of Politics, 2016

- **Puzzle:** the 1994 reform reduced same-party competition, but later studies still described campaigns as pork-oriented.
- **Question:** did candidates shift from benefits for small groups to broader public policies?
- **Evidence:** 7,497 candidate manifestos from eight elections, 1986–2009.

Reference: Catalinac (2016)

Manifestos become campaign-priority measures

Case 1 · How LDA is used · 1/3

Research step	What Catalinac does
Discover topics	LDA identifies 69 recurring themes in 7,497 manifestos
Read and check	Inspect 15 top words and 10 representative manifestos per topic
Define concepts	Use who benefits to classify 14 pork and 52 policy topics
Measure candidates	Sum each category's topic shares within every manifesto
Test the puzzle	Compare those shares across elections and levels of same-party competition

LDA finds recurring themes; close reading gives them political meaning.

Reference: Catalinac (2016)

Manifestos become campaign-priority measures

Case 1 · How LDA is used · 2/3

Research step	What Catalinac does
Discover topics	LDA identifies 69 recurring themes in 7,497 manifestos
Read and check	Inspect 15 top words and 10 representative manifestos per topic
Define concepts	Use who benefits to classify 14 pork and 52 policy topics
Measure candidates	Sum each category's topic shares within every manifesto
Test the puzzle	Compare those shares across elections and levels of same-party competition

Catalinac classifies interpreted topics by who benefits, creating pork and policy measures.

Reference: Catalinac (2016)

Manifestos become campaign-priority measures

Case 1 · How LDA is used · 3/3

Research step	What Catalinac does
Discover topics	LDA identifies 69 recurring themes in 7,497 manifestos
Read and check	Inspect 15 top words and 10 representative manifestos per topic
Define concepts	Use who benefits to classify 14 pork and 52 policy topics
Measure candidates	Sum each category's topic shares within every manifesto
Test the puzzle	Compare those shares across elections and levels of same-party competition

↓

Candidate-level shares are compared across elections and levels of same-party competition.

Reference: Catalinac (2016)

Pork and policy across Japanese elections

Case 1 · The measure enters the result

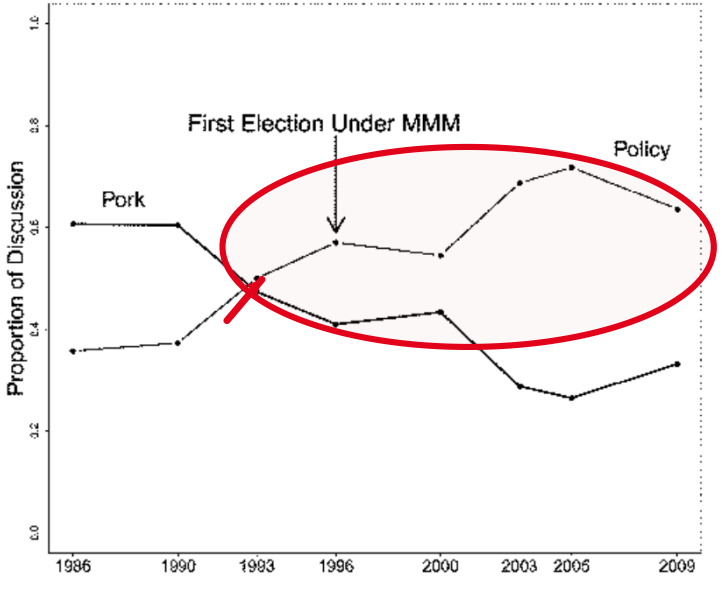


Figure 1. LDP candidates switched to more policy and less pork in the 1993 election and continued with this strategy under MMM. This figure plots the mean proportions of discussion devoted to pork and policy, respectively, in the 2,355 manifestos produced by LDP candidates in these eight elections.

Start with the red circle: **broad policy promises** rise before the new system is first used in 1996.

Reference: Catalinac (2016)

Pork and policy across Japanese elections

Case 1 · The measure enters the result

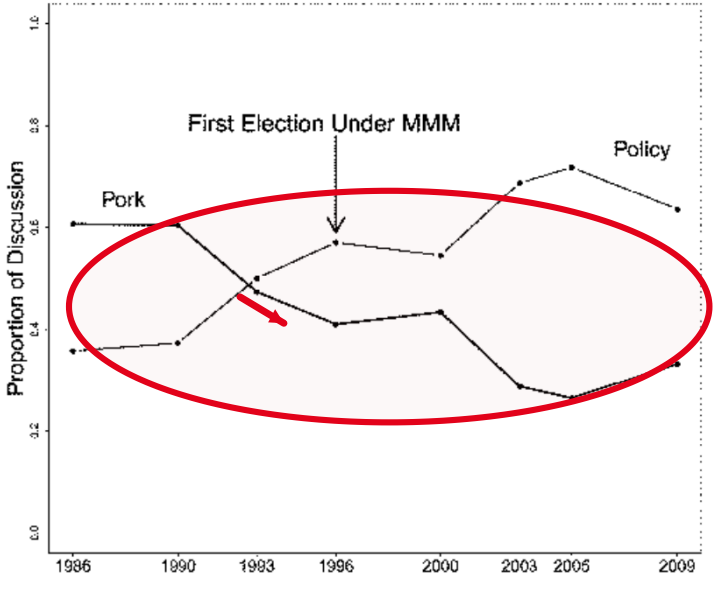


Figure 1. LDP candidates switched to more policy and less pork in the 1993 election and continued with this strategy under MMM. This figure plots the mean proportions of discussion devoted to pork and policy, respectively, in the 2,355 manifestos produced by LDP candidates in these eight elections.

Now follow the lower line: **local-benefit promises** fall while broad policy promises rise.

Reference: Catalinac (2016)

Competition and campaign priorities

Case 1 · A complementary comparison

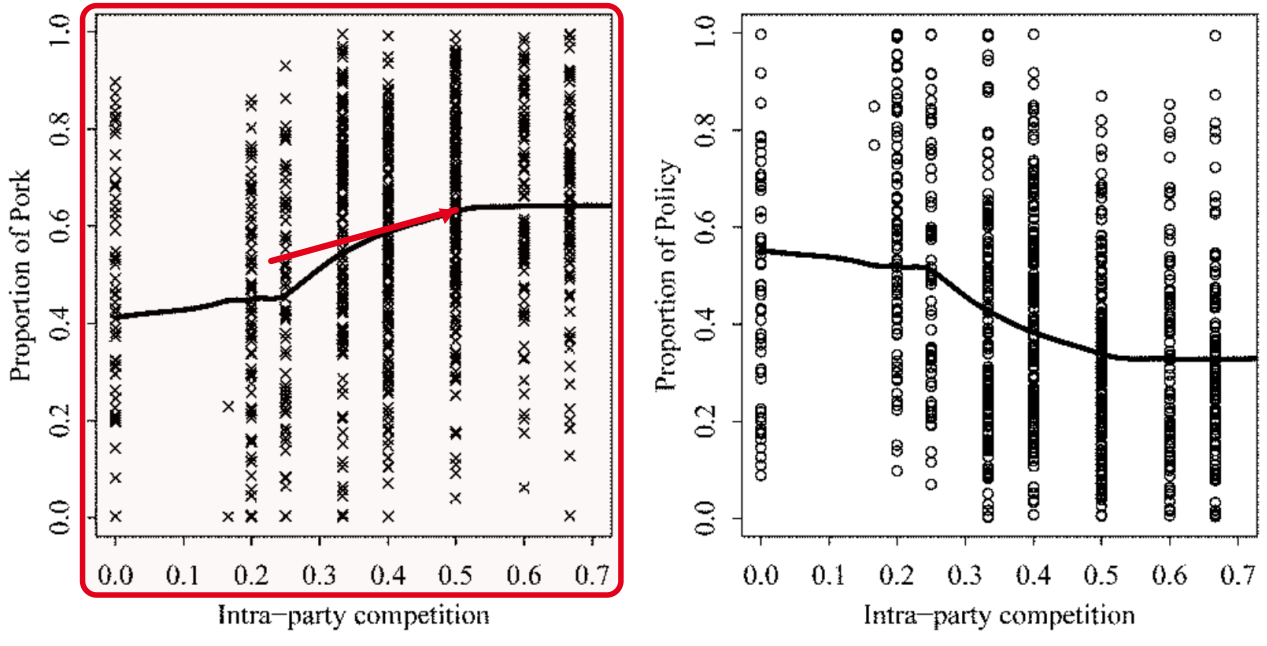


Figure 3. LDP candidates facing higher levels of intraparty competition discussed more pork and less policy than LDP candidates facing lower levels of intraparty competition under SNTV-MMD, where level of intraparty competition is measured by number of LDP opponents relative to district magnitude. The left panel shows that discussion of pork is higher at higher levels of intraparty competition. The right panel shows that discussion of policy is lower at higher levels of intraparty competition.

Left panel: more **same-party rivals** are linked to more local-benefit promises.

Reference: Catalinac (2016)

Competition and campaign priorities

Case 1 · A complementary comparison

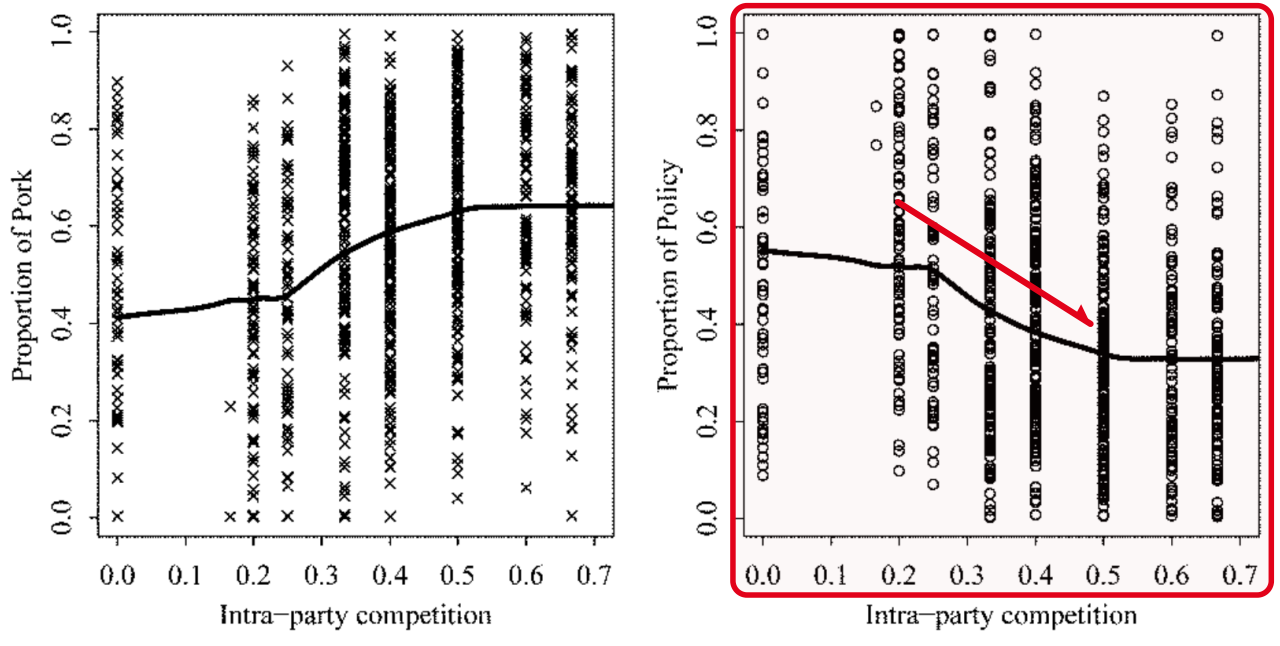


Figure 3. LDP candidates facing higher levels of intraparty competition discussed more pork and less policy than LDP candidates facing lower levels of intraparty competition under SNTV-MMD, where level of intraparty competition is measured by number of LDP opponents relative to district magnitude. The left panel shows that discussion of pork is higher at higher levels of intraparty competition. The right panel shows that discussion of policy is lower at higher levels of intraparty competition.

Right panel: more **same-party rivals** are linked to fewer broad policy promises.

Reference: Catalinac (2016)

Opposition candidates emphasize policy

Case 1 · Opposition comparison

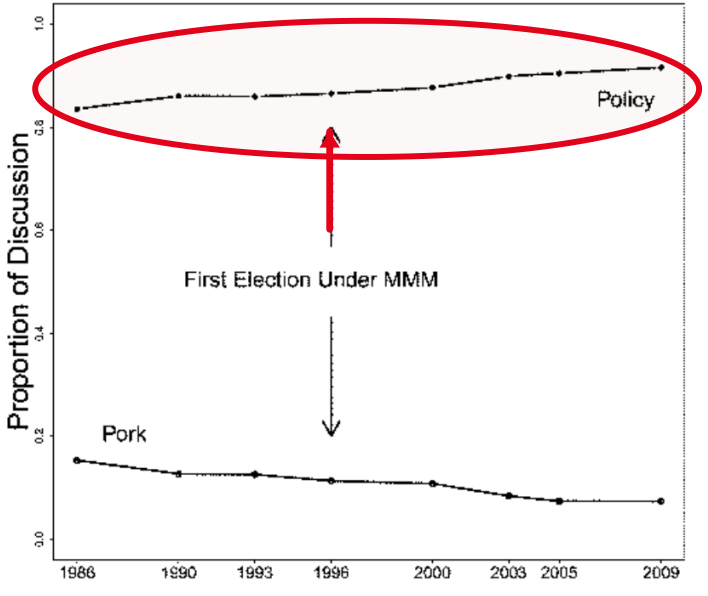


Figure 4. Opposition candidates adopted electoral strategies of more policy and less pork in elections held under both SNTV-MMD and MMM. This figure plots the mean proportions of discussion comprised of pork and policy in the 4,505 manifestos produced by opposition candidates running from the 14 major opposition parties in these eight elections.

The opposition already makes many **broad policy promises** before 1996, and the line stays high.

Reference: Catalinac (2016)

Opposition candidates emphasize policy

Case 1 · Opposition comparison

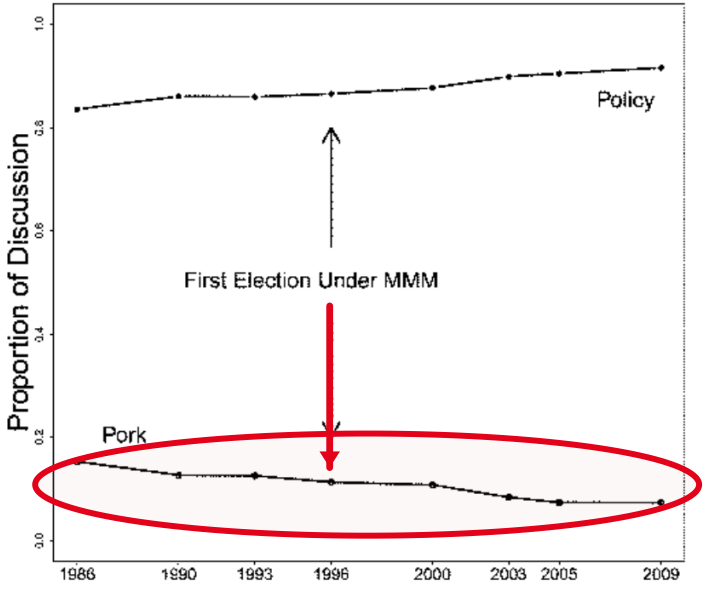


Figure 4. Opposition candidates adopted electoral strategies of more policy and less pork in elections held under both SNTV-MMD and MMM. This figure plots the mean proportions of discussion comprised of pork and policy in the 4,505 manifestos produced by opposition candidates running from the 14 major opposition parties in these eight elections.

The opposition's **local-benefit promises** stay low. This observational comparison does not identify a causal effect.

Reference: Catalinac (2016)

Policy diffusion before adoption

Case 2 · Issue definition



Fabrizio Gilardi, Charles R. Shipan and Bruno Wüest

American Journal of Political Science, 2021

- **Puzzle:** diffusion research usually observes policy adoption after the debate has already begun.
- **Question:** do earlier smoking bans elsewhere change how a state's newspapers define the issue?
- **Evidence:** 52,675 paragraphs from 49 state newspapers, 1996–2013.

Reference: Gilardi, Shipan & Wüest (2021)

News paragraphs become policy frames

Case 2 · How STM is used · 1/3

Research step	What the researchers do
Collect reporting	Search 49 state newspapers, 1996–2013
Keep relevant paragraphs	Use crowd labels to train an SVM relevance filter
Discover frames	Fit a 12-topic STM to 52,675 paragraphs
Measure issue definition	Estimate how common each frame is
Study diffusion	Relate frame prevalence to earlier adoptions in other states

First collect possible reports, then use human labels to keep relevant paragraphs.

Reference: Gilardi, Shipan & Wüest (2021)

News paragraphs become policy frames

Case 2 · How STM is used · 2/3

Research step	What the researchers do
Collect reporting	Search 49 state newspapers, 1996–2013
Keep relevant paragraphs	Use crowd labels to train an SVM relevance filter
Discover frames	Fit a 12-topic STM to 52,675 paragraphs
Measure issue definition	Estimate how common each frame is
Study diffusion	Relate frame prevalence to earlier adoptions in other states

STM finds recurring frames; topic prevalence measures how common each frame is.

Reference: Gilardi, Shipan & Wüest (2021)

News paragraphs become policy frames

Case 2 · How STM is used · 3/3

Research step	What the researchers do
Collect reporting	Search 49 state newspapers, 1996–2013
Keep relevant paragraphs	Use crowd labels to train an SVM relevance filter
Discover frames	Fit a 12-topic STM to 52,675 paragraphs
Measure issue definition	Estimate how common each frame is
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↓

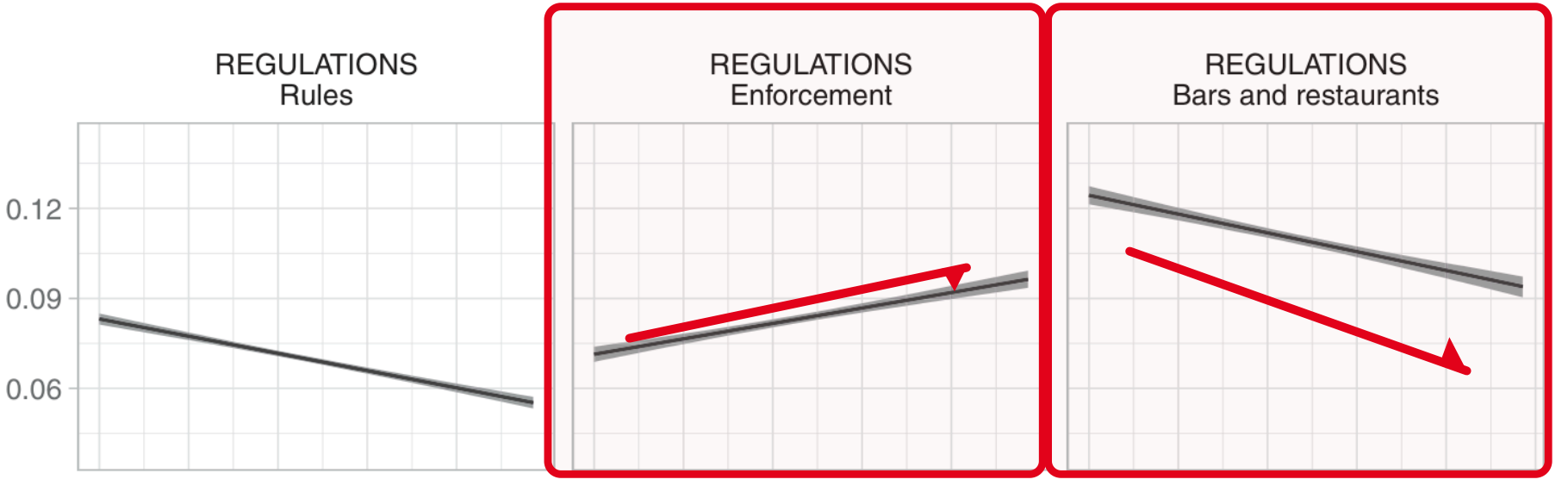
The analysis relates frame prevalence to earlier policy adoptions in connected states.

Reference: Gilardi, Shipan & Wüest (2021)

Practical policy frames change

Case 2 · Learning from other states

FIGURE 2 Topic Prevalence and the Share of Prior Adoptions within a State's Diffusion Network

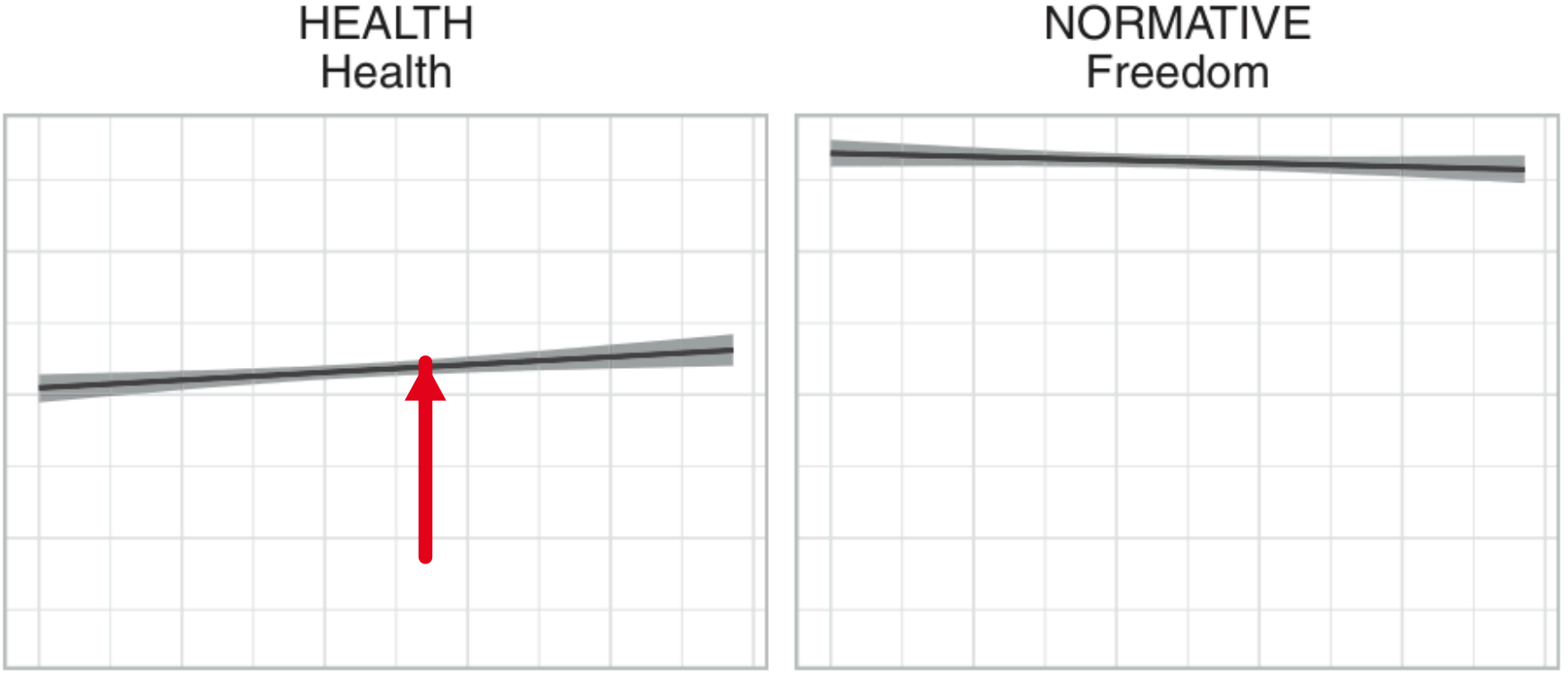


Moving right means more related states have adopted a ban. Restaurant concerns fall, while enforcement becomes more prominent.

Reference: Gilardi, Shipan & Wüest (2021)

Some frames remain stable

Case 2 · Stable frame: Health

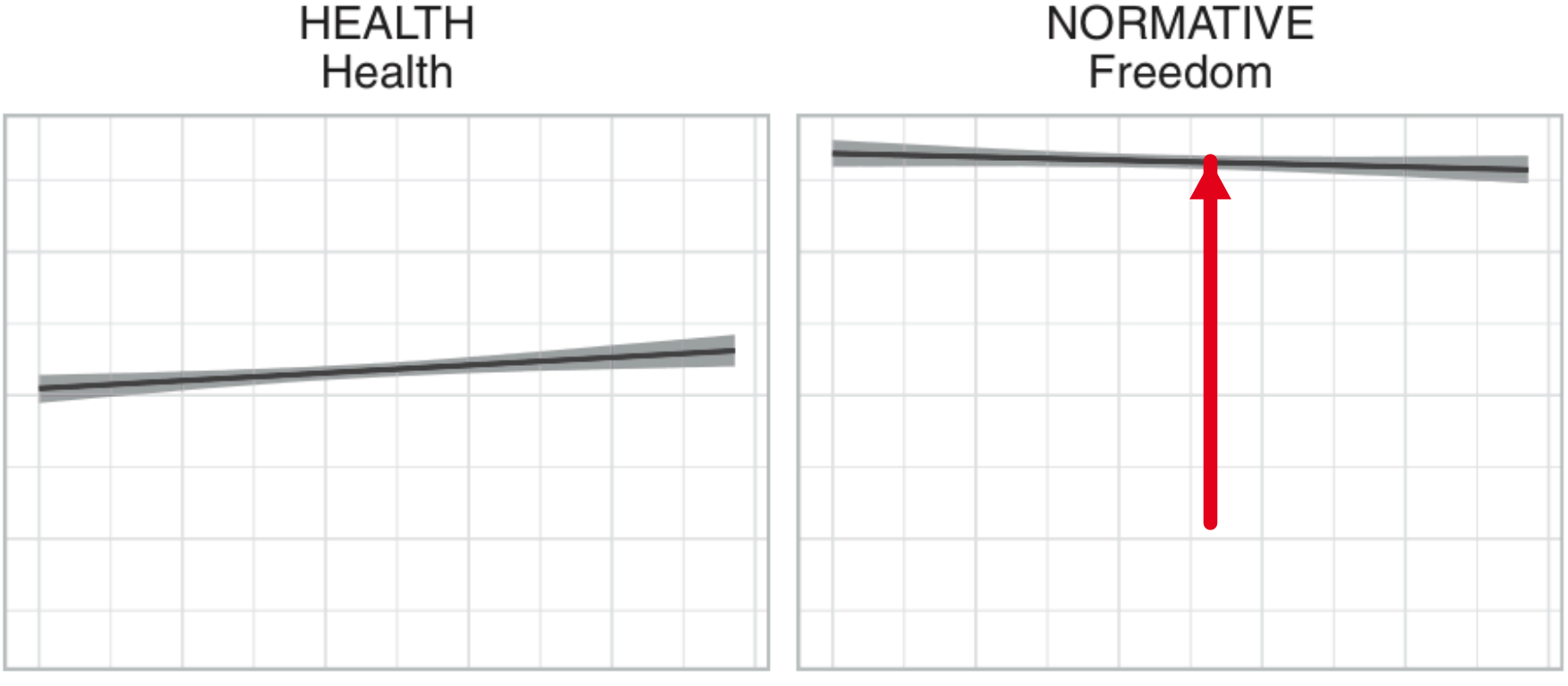


Health changes little as prior adoption rises.

Reference: Gilardi, Shipan & Wüest (2021)

Some frames remain stable

Case 2 · Stable frame: Freedom

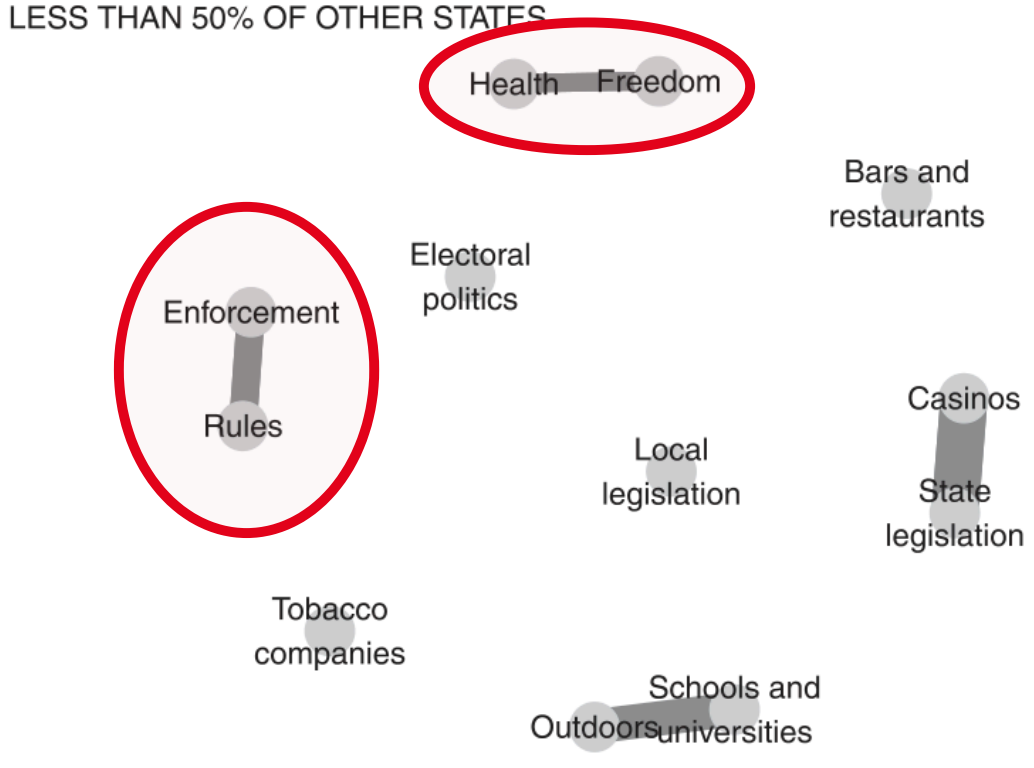


Freedom also stays nearly flat across adoption levels.

Reference: Gilardi, Shipan & Wüest (2021)

Frames remain separate at lower adoption

Case 2 · Optional detail · lower adoption

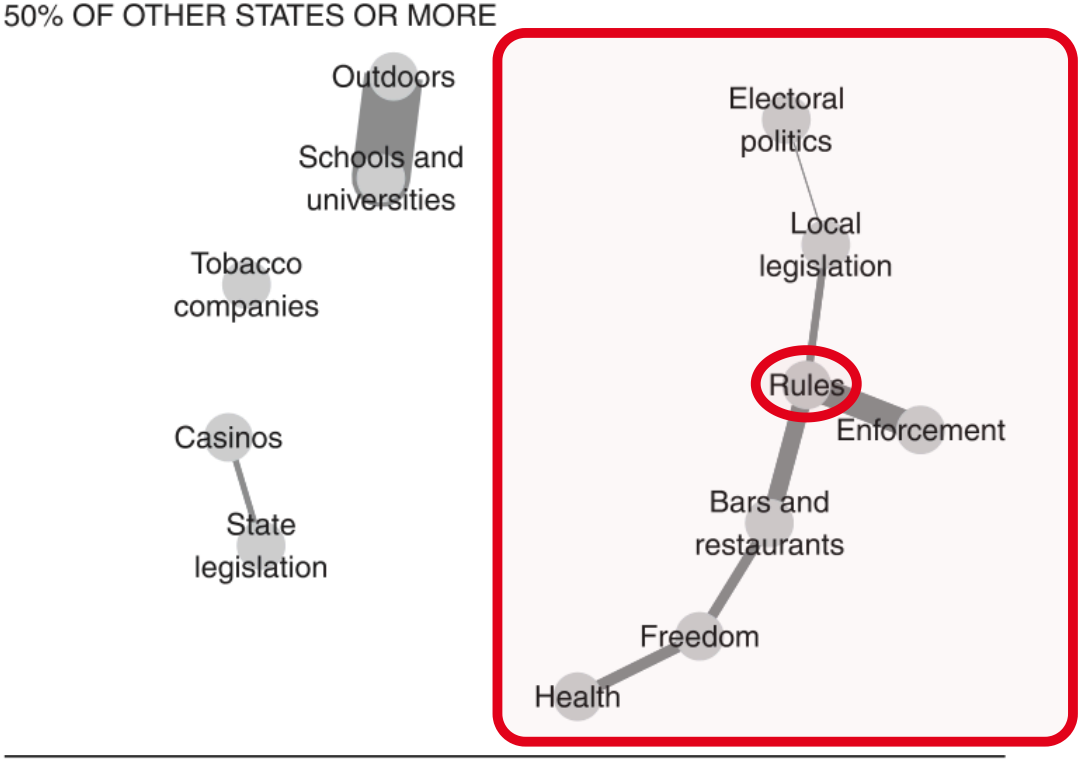


At lower adoption, the Rules topic has few strong correlations with the other frames.

Reference: Gilardi, Shipan & Wüest (2021)

A broader frame emerges at higher adoption

Case 2 · Optional detail · higher adoption



At higher adoption, Rules is correlated with several practical, value and political frames.

Reference: Gilardi, Shipan & Wüest (2021)

Who leads? Who follows?

Case 3 · Social media and issue attention



American Political Science Review

Article contents

Who Leads? Who Follows? Measuring Issue Attention and Agenda Setting by Legislators and the Mass Public Using Social Media Data

Published online by Cambridge University Press: **12 July 2019**

PABLO BARBERÁ, ANDREU CASAS, JONATHAN NAGLER, PATRICK J. EGAN, RICHARD BONNEAU, JOHN T. JOST and JOSHUA A. TUCKER

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Pablo Barberá and colleagues

American Political Science Review, 2019

- **Question:** do politicians lead public attention, or follow it?
- **Texts:** Congress, public and media tweets, 2013–2014.
- **Use:** compare daily attention to shared issues.

Reference: Barberá et al. (2019)

Tweets become daily issue-attention measures

Case 3 · How the topic model is used · 1/3

Research step	What the researchers do
Define documents	Combine congressional tweets by day, party and chamber
Discover shared themes	Fit a 100-topic LDA to the grouped congressional texts
Interpret issues	Read topic words and merge related political themes
Construct measures	Estimate daily issue attention for politicians, public samples and media
Analyse timing	Compare whose earlier attention predicts another group's later attention

Tweets are grouped into comparable daily documents before LDA finds recurring themes.

Reference: Barberá et al. (2019)

Tweets become daily issue-attention measures

Case 3 · How the topic model is used · 2/3

Research step	What the researchers do
Define documents	Combine congressional tweets by day, party and chamber
Discover shared themes	Fit a 100-topic LDA to the grouped congressional texts
Interpret issues	Read topic words and merge related political themes
Construct measures	Estimate daily issue attention for politicians, public samples and media
Analyse timing	Compare whose earlier attention predicts another group's later attention

Researchers turn interpreted topics into comparable daily attention measures.

Reference: Barberá et al. (2019)

Tweets become daily issue-attention measures

Case 3 · How the topic model is used · 3/3

Research step	What the researchers do
Define documents	Combine congressional tweets by day, party and chamber
Discover shared themes	Fit a 100-topic LDA to the grouped congressional texts
Interpret issues	Read topic words and merge related political themes
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↓

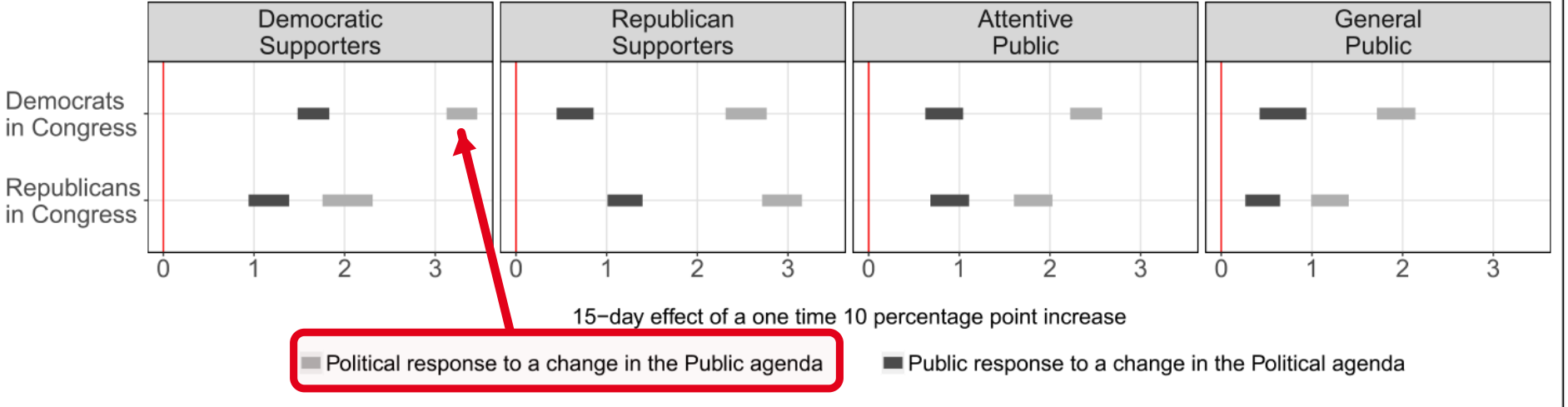
Daily attention enters a time-series model that compares who moves first.

Reference: Barberá et al. (2019)

Who responds to whose attention?

Case 3 · Direction one

FIGURE 3. Politicians' Ability to Set Public Agendas versus the Ability of the Public to Influence Political Agendas



Note: The coefficients (with 95% confidence intervals) indicate (in percentage points) how much more cumulative attention the groups in the panel titles paid to a given issue as a result of the groups in the *y*-axis increasing the attention to the same issue by 10 percentage points (in black) 15 days ago. The gray coefficients indicate the vice versa effect.

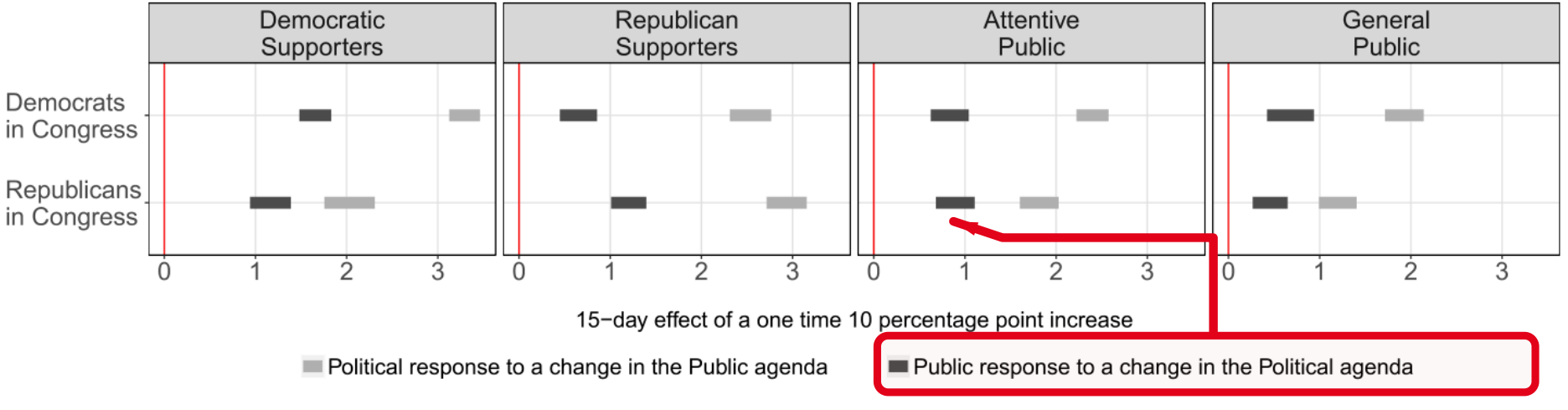
Gray: when public attention rises, politicians often pay more attention later.

Reference: Barberá et al. (2019)

Who responds to whose attention?

Case 3 · Reverse direction and caveat

FIGURE 3. Politicians' Ability to Set Public Agendas versus the Ability of the Public to Influence Political Agendas



Note: The coefficients (with 95% confidence intervals) indicate (in percentage points) how much more cumulative attention the groups in the panel titles paid to a given issue as a result of the groups in the *y*-axis increasing the attention to the same issue by 10 percentage points (in black) 15 days ago. The gray coefficients indicate the vice versa effect.

Black: when politicians' attention rises, the public may pay more attention later.

Reference: Barberá et al. (2019)

Immigration and shared democratic values

Case 4 · Topic models for theory building



Political Behavior (2024) 46:1887–1911
https://doi.org/10.1007/s11109-023-09900-y

ORIGINAL PAPER

Check for updates

Birds of a Feather: Sharing Democratic Values Eases Immigration in a Postmaterialist Society

Jin Jiang[1] · Shouzhi Xia[2] · Dong Zhang[3]

Jin Jiang, Shouzhi Xia and Dong Zhang

Political Behavior, 2024

- **Puzzle:** research emphasizes economic and cultural concerns more than shared political values.
- **Question:** can shared democratic values reduce support for restrictive immigration policy?
- **Evidence:** researchers use the STM results to develop the hypothesis. A preregistered experiment tests it.

Reference: Jiang, Xia & Zhang (2024)

From news topics to a hypothesis

Case 4 · How STM supports theory building · 1/3

Research step	Use in this article
Discover concerns	STM identifies five topics in 4,453 news articles
Read and label	Inspect characteristic words and representative news
Compare over time	Estimate changes in topic prevalence, 2003–2020
Build a hypothesis	Rising political attention motivates a shared-values argument
Test separately	Randomize messages in a preregistered experiment

STM finds recurring concerns; researchers read the texts and supply the labels.

Reference: Jiang, Xia & Zhang (2024)

From news topics to a hypothesis

Case 4 · How STM supports theory building · 2/3

Research step	Use in this article
Discover concerns	STM identifies five topics in 4,453 news articles
Read and label	Inspect characteristic words and representative news
Compare over time	Estimate changes in topic prevalence, 2003–2020
↓ Build a hypothesis	Rising political attention motivates a shared-values argument
Test separately	Randomize messages in a preregistered experiment

Changes in topic attention motivate the shared-democratic-values hypothesis.

Reference: Jiang, Xia & Zhang (2024)

From news topics to a hypothesis

Case 4 · How STM supports theory building · 3/3

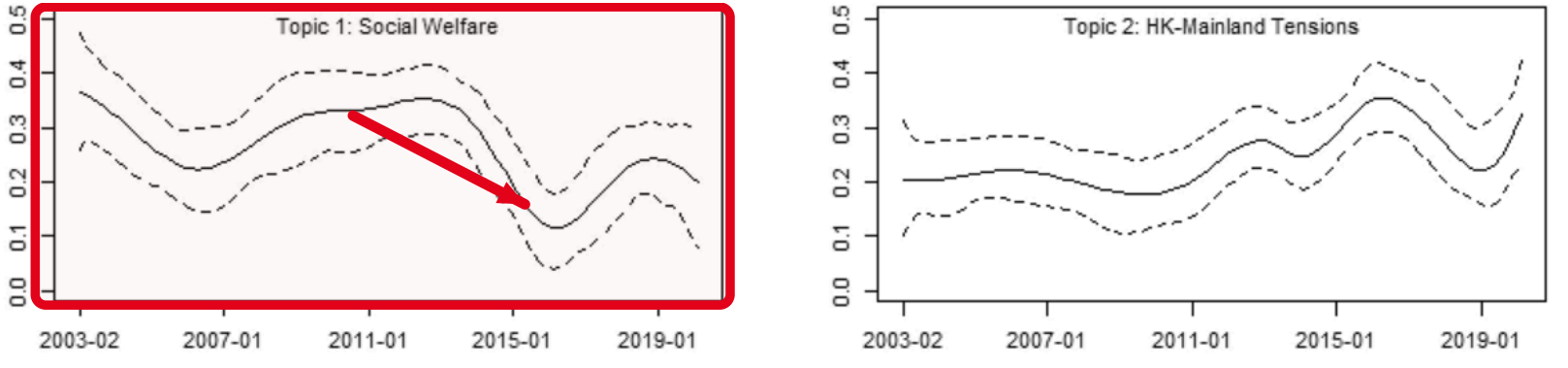
Research step	Use in this article
Discover concerns	STM identifies five topics in 4,453 news articles
Read and label	Inspect characteristic words and representative news
Compare over time	Estimate changes in topic prevalence, 2003–2020
Build a hypothesis	Rising political attention motivates a shared-values argument
↓ Test separately	Randomize messages in a preregistered experiment

Researchers use the STM results to develop the hypothesis. Random assignment tests the message effect.

Reference: Jiang, Xia & Zhang (2024)

Changing attention to mainland immigration

Case 4 · Theory-building evidence

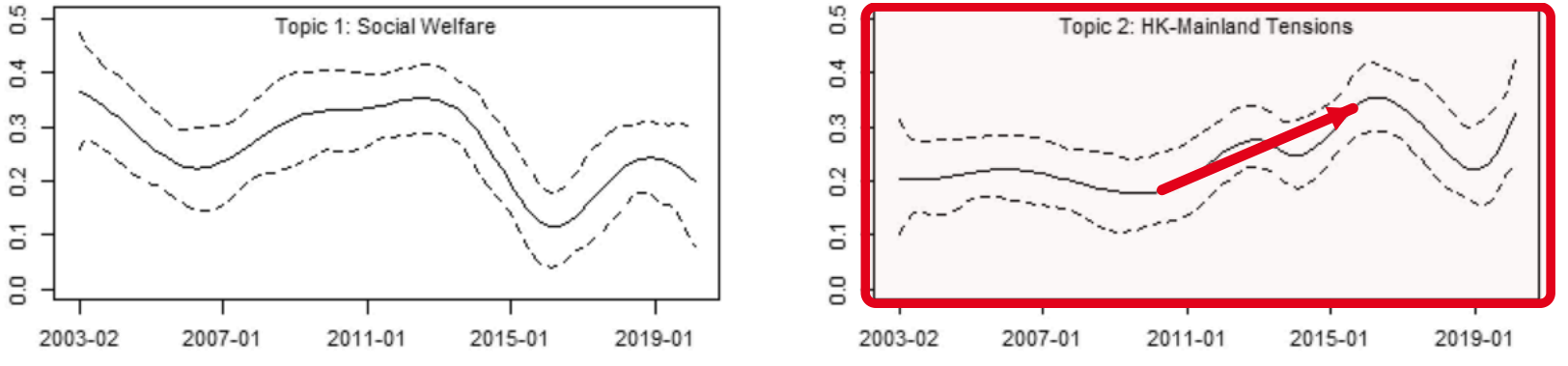


News attention to social welfare falls after the mid-2010s. Dashed lines show 95% uncertainty.

Reference: Jiang, Xia & Zhang (2024)

Changing attention to mainland immigration

Case 4 · Hypothesis development



Attention to HK–Mainland tensions rises. This pattern motivates a shared-values hypothesis.

Reference: Jiang, Xia & Zhang (2024)

National problems in respondents' own words

Case 5 · Public issue priorities

Structural Topic Models for Open-Ended Survey Responses

Margaret E. Roberts University of California, San Diego
Brandon M. Stewart Harvard University
Dustin Tingley Harvard University
Christopher Lucas Harvard University
Jetson Leder-Luis California Institute of Technology
Shana Kushner Gadarian Syracuse University
Bethany Albertson University of Texas at Austin
David G. Rand Yale University

Margaret E. Roberts and colleagues

American Journal of Political Science, 2014

- **Question:** who names war as a national problem?
- **Document:** one ANES open-ended answer.
- **Comparison:** party identification and education.

Reference: Roberts et al. (2014)

Survey answers become issue comparisons

Case 5 · How the model is used

Step	Role in the research design
Texts	Answers to the most-important-national-problem question
Model inputs	60-topic STM; party, education, age and party × education
Output used	Estimated war-related share in each answer
Political comparison	War-topic relationship with party and education
Check	Compare with ANES human-coded war categories

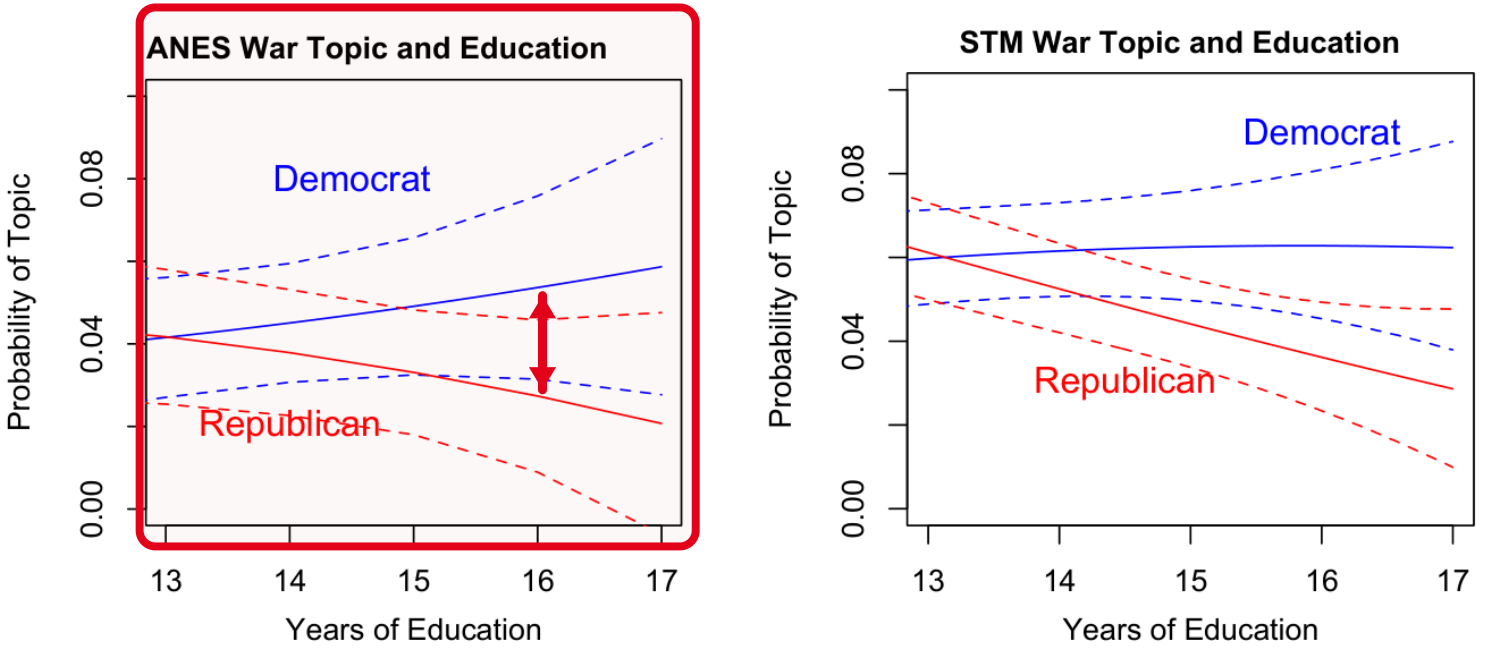
Human category codes provide a benchmark; they do not train this STM.

Reference: Roberts et al. (2014)

Survey priorities and human-coded evidence

Case 5 · Human-coded benchmark

FIGURE 18 Comparison of Covariate Relationships



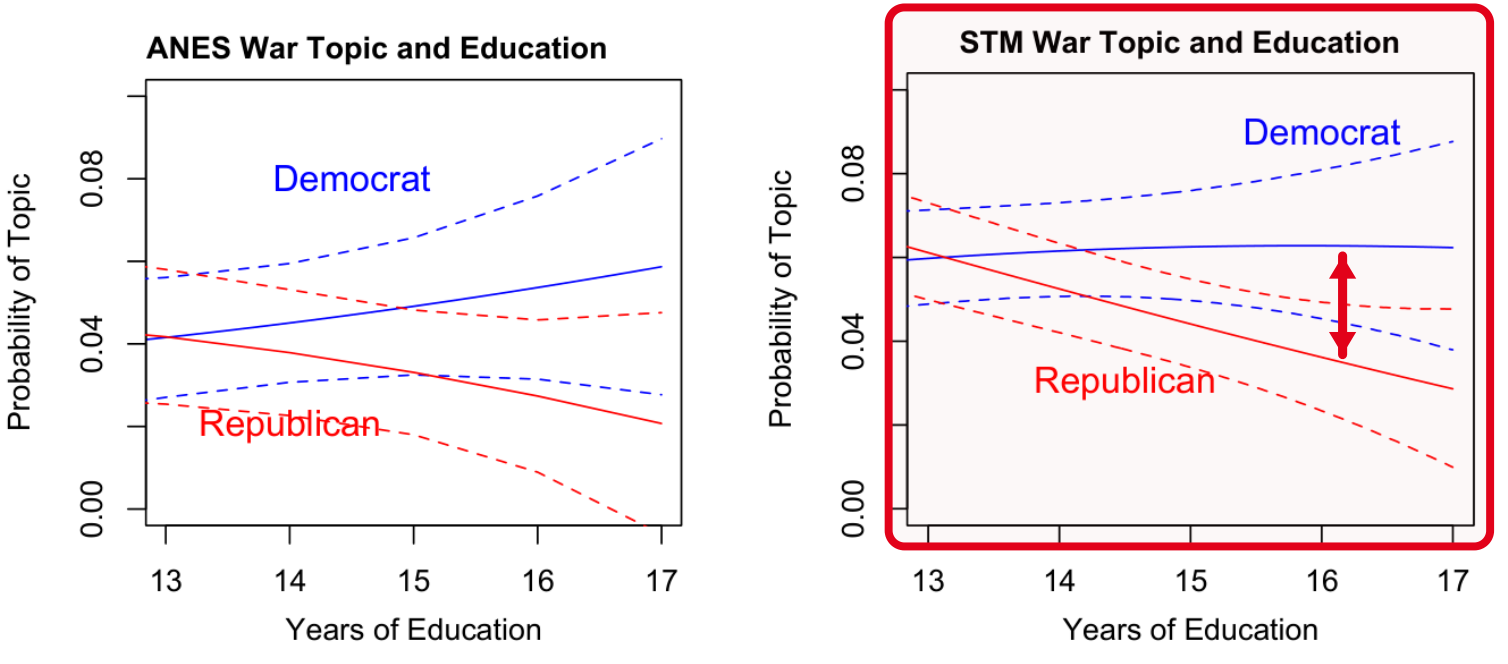
Left: the human-coded party gap is larger at higher education.

Reference: Roberts et al. (2014)

Survey priorities and human-coded evidence

Case 5 · STM comparison

FIGURE 18 Comparison of Covariate Relationships



Right: the STM also shows a wider party gap; the Democratic line is roughly flat.

Reference: Roberts et al. (2014)

Predicting conflict from newspaper topics

Case 6 · Hannes Mueller and Christopher Rauh · APSR 2018

Puzzle: Slow-moving country indicators often show *where* conflict is common, but miss *when* risk rises. Can news language improve one-year-ahead forecasts?

Step	What they do
Corpus	More than 700,000 Economist, New York Times and Washington Post articles
Measure	Article-level LDA, K = 15, aggregated to country-year topic shares
Forecast	Re-estimate with text through year T; predict conflict onset at T + 1
Evaluate	Repeat for 1996–2014; read ROC together with precision–recall

What this supports: predictive usefulness. Labels and causal claims need separate evidence.

Reference: Mueller & Rauh (2018)

From topics to political evidence

Study	Model contribution	Next research step
Catalinac	Topic shares	Define pork/policy; compare candidates
Gilardi et al.	Frame proportions	Relate frames to prior policy adoption
Barberá et al.	Daily attention	Compare who moves first
Jiang et al.	Topic trends	Develop a hypothesis; test separately
Roberts et al.	Survey topic shares	Compare patterns with human coding
Mueller & Rauh	Country-year topic shares	Test next-year conflict forecasts

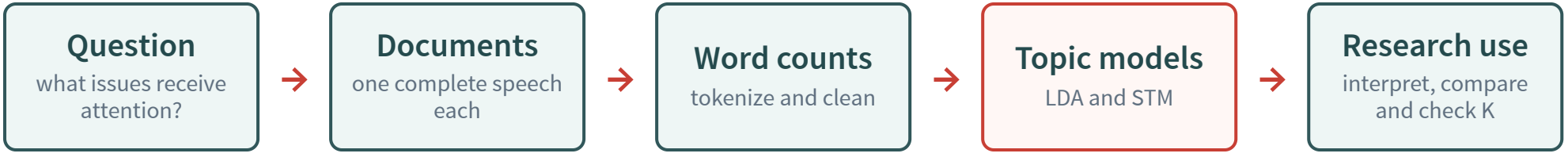
The research design determines what the evidence supports.

Reference: Catalinac (2016); Gilardi, Shipan & Wüest (2021); Barberá et al. (2019); Jiang, Xia & Zhang (2024); Roberts et al. (2014); Mueller & Rauh (2018)

Applying topic models to political speeches

Teacher demonstration · students repeat it in the Lab

Political question: Which issues do party leaders emphasize in their speeches?



We will explore themes with LDA, add document information with STM, then compare topic resolutions using perplexity and original texts.

One workflow: fit, interpret, compare and validate.

Data source: Essex Summer School 2026, Day 7; British House of Commons party-leader speeches. W8 fixed teaching sample: 719 speeches.

Step 1: Read the speeches

LDA application · 1/6

```r
library(quanteda)

data_dir <- "analysis/ess_lab"
topic_docs <- readRDS(file.path(
  data_dir, "sample_raw.rds"))

topic_corpus <- corpus(
  topic_docs,
  text_field = "text",
  docid_field = "doc_id")
```

Corpus = a collection of texts

719 complete speeches

The example used on all six pages

Field	Value
ID	`ess_hoc_00004`
Leader	Margaret Thatcher
Party	Conservative
Date	1979-06-26

Here, one document means one complete speech. The corpus keeps each speech together with its ID and background information.

Data source: Essex Summer School 2026, Day 7, British House of Commons party-leader speeches (p. 3). W8 fixed teaching sample: 719 speeches.

Step 2: Split one speech into words

LDA application · 2/6

```r
id <- "ess_hoc_00004"

raw_tokens <- tokens(
  topic_corpus[id],
  what = "word")

raw_tokens[[id]][1:21]
```

Original sentence

> “With permission, Mr. Speaker I should like to make a statement on the European Council Meeting at Strasbourg.”

First 21 raw tokens

```
“With” “permission” “,” “Mr” “.” “Speaker”
“I” “should” “like” “to” “make” “a”
“statement”
“on” “the” “European” “Council” “Meeting”
“at” “Strasbourg” “.”
```

Tokenization splits a speech into separate units. At this stage, punctuation and common words can still appear.

Data source: Essex Summer School 2026, Day 7; British House of Commons party-leader speeches. Example: speech `ess_hoc_00004`.

Step 3: Remove words that do not help

LDA application · 3/6

```
clean_tokens <- tokens(
  topic_corpus,
  remove_punct = TRUE,
  remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(c(
    stopwords("en"),
    "mr", "speaker"))

clean_tokens[[id]][1:8]
```

First eight cleaned tokens

“permission” “like” “make” “statement” “european” “council” “meeting” “strasbourg”

Remove punctuation
Use lowercase
Remove very common words

Cleaning keeps the words that the model will compare. Every removal rule needs a reason.

Data source: Essex Summer School 2026, Day 7; British House of Commons party-leader speeches. W8 fixed teaching sample: 719 speeches.

Step 4: Fit the LDA model

LDA application · 4/6

Full DFM after cleaning: 719 × 2,831

```r
# Use the DFM constructed in Lab R01c.
set.seed(20260710)
lda_6 <- seededlda::textmodel_lda(
  topic_dfm,
  k = 6,
  alpha = .5, beta = .1,
  max_iter = 500,
  auto_iter = FALSE,
  batch_size = 1,
  verbose = FALSE)
```

What the fitted model stores

Output	Question it answers
Topic–word probabilities	Which words characterize each topic?
Document–topic proportions	Which topics appear in each speech?

A speech can mix topics; each topic favors different words. LDA estimates both from word patterns across the corpus.

This fit uses the full DFM built in the Lab. The short cleaning example on the previous page is for illustration.

Data source: Essex Summer School 2026, Day 7, British House of Commons party-leader speeches (p. 21). seededlda documentation.

Step 5: Topic words

LDA application · 5/6

`terms(lda_6, 5)` shows the five highest-probability words in each topic.

Topic	Highest-probability words	Provisional reading
T1	european, council, community, countries, agreed	European Council negotiations
T2	government, state, made, members, service	Government and public service?
T3	prime, people, government, said, can	General political leadership?
T4	government, people, unemployment, economy, year	Economy and unemployment
T5	people, can, country, want, support	Public appeals and support?
T6	united, must, international, security, nations	International security

Topic words suggest a provisional label. Original speeches show whether the label holds.

Data source: W8 six-topic LDA fitted to the fixed sample of 719 British House of Commons party-leader speeches. Student Lab R02a.

Step 6: One document's topics

LDA application · 6/6

```r
topic_names <- c(
  "European Council negotiations",
  "Government and public service?",
  "General political leadership?",
  "Economy and unemployment",
  "Public appeals and support?",
  "International security")

id <- "ess_hoc_00004"
i <- match(id, topic_docs$doc_id)

doc_topics <- lda_6$theta[i, ]
round(doc_topics, 3)

dominant_topic <- which.max(doc_topics)
topic_names[dominant_topic]
```

Document `ess_hoc_00004`

T1	T2	T3	T4	T5	T6
.874	.067	.012	.002	.012	.032

Largest share: T1
Provisional name: European Council negotiations

> "…a statement on the European Council Meeting at Strasbourg."

The document keeps all six proportions. The dominant topic is a short summary of its largest share.

Data source: W8 six-topic LDA and original speech `ess_hoc_00004`. Student Lab R02b–R03b.

Using STM to compare party attention

Question: do Conservative and Labour leaders emphasize the same issues?

Keep from the LDA example	Add for the STM example
The same 719 complete speeches	Party and election period for each speech
The same word-count matrix	A model of how topic shares vary with this information

We will follow the same Thatcher speech, then compare one topic across parties.

Focus: topic prevalence — how much a topic is discussed.

Reference: Roberts et al. (2014)

STM Step 1: Add document information

STM application · 1/5 · Teacher demonstration

```r
metadata <- data.frame(
  doc_id = topic_docs$doc_id,
  party_family = topic_docs$party_family,
  period = topic_docs$period)
i <- match("ess_hoc_00004",
metadata$doc_id)
metadata[i, c("party_family", "period")]
```

The same speech: `ess_hoc_00004`

Field	Stored value
Party	Conservative
Election period	47th Parliament (1979-1983)

719 metadata rows, in the same order as the 719 speech documents.

`period` is a category, not a numeric time trend.

One speech, one metadata row. Party and period now accompany the word counts.

W8 teacher demonstration. Same 719 speeches and STM specification as Lab R05.

STM Step 2: Specify the model

STM application · 2/5 · Teacher demonstration

Fit once · reference specification

```
stm_input <- convert(topic_dfm, to = "stm")
stm_6 <- stm::stm(
  documents = stm_input$documents,
  vocab = stm_input$vocab, K = 6,
  prevalence = ~ party_family + period,
  data = metadata, seed = 20260710,
  init.type = "Spectral", max.em.its = 100,
  emtol = 1e-4, verbose = FALSE)
```

Read the fitted result

```
dim(stm_6$theta)
```

Output: 719 documents × 6 topics.

`prevalence = ~ party_family + period`

Party and period help model topic shares. Word choice within topics is shared in this example.

Metadata enters topic estimation. R05a fits this specification in the Lab.

W8 teacher demonstration. Same 719 speeches and STM specification as Lab R05. stm reference.

STM Step 3: Read one topic

STM application · 3/5 · Teacher demonstration

```
stm::labelTopics(
  stm_6, topics = 4, n = 5)

round(stm_6$theta[i, 4], 3)
```

STM Topic 4

Word summary	Five characteristic words
Probability	council, community, agreed, united, european
FREX	argentine, commonwealth, nations, agreed, meeting

Provisional label: International affairs?

The same Thatcher speech has **97.9%** in this STM topic.

Read words and original text together. Topic numbers belong to their own fitted model.

W8 teacher demonstration. Same 719 speeches and STM specification as Lab R05. Selected probability/FREX words from R05a; original speech ess_hoc_00004.

STM Step 4: Compare party averages

STM application · 4/5 · Teacher demonstration

```r
party_means <- tapply(
  stm_6$theta[, 4],
  metadata$party_family, mean)

round(100 * party_means, 2)
```

Mean share of STM Topic 4

Party	Mean topic share
Conservative	13.66%
Labour	7.63%

Raw Labour − Conservative: -6.03 percentage points.

Each sampled speech has equal weight.

These are descriptive averages. The speeches come from different election periods.

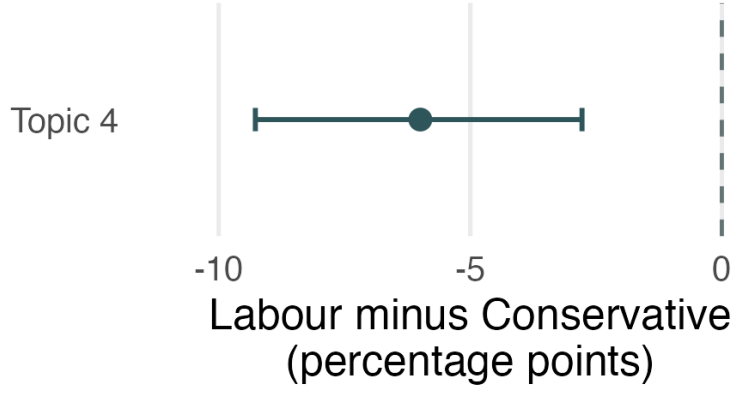
W8 teacher demonstration. Same 719 speeches and STM specification as Lab R05. Same unadjusted means as R05b.

STM Step 5: Account for election period

STM application · 5/5 · Teacher demonstration

```r
set.seed(20260710)
party_effects <- stm::estimateEffect(
  1:6 ~ party_family + period,
  stm_6, metadata, uncertainty = "Global",
nsims = 50)
set.seed(20260710)
z <- plot(party_effects, covariate =
"party_family",
  method = "difference", cov.value1 =
"Labour",
  cov.value2 = "Conservative", topics =
1:6,
  nsims = 1000, omit.plot = TRUE,
labeltype = "numbers")
round(100 * c(z$means[[4]], z$cis[[4]]),
2)
```

Adjusted comparison · Topic 4



Quantity	Percentage points
Labour − Conservative	-5.99
95% interval	[-9.27, -2.78]

The comparison accounts for election period. It describes a conditional association.

W8 teacher demonstration. Same 719 speeches and STM specification as Lab R05. Exact R05c–R05d conditional difference; plot selects Topic 4 only.

Assumptions and document boundaries

Modelling choice	Question for the researcher
Bag of words	Does word order carry information we need?
A common set of topics	Are language and topic meanings comparable across the corpus?
Document boundaries	Does one speech contain enough context for a mixture?
Vocabulary selection	Which distinctions survive cleaning and trimming?
K	Which issues split or merge when the number of topics changes?
Alpha and initialization	Are mixtures and results stable across reasonable settings?

These choices shape the topics before researchers name or compare them.

Reference: Blei, Ng & Jordan (2003); ESS course; Grimmer, Roberts & Stewart (2022)

Topic validation asks different questions

Check	Question
Held-out word fit	Can the model account for unseen words?
Words and documents	Does the label describe the content?
Coherence and exclusivity	Are prominent words related and distinctive?
Sensitivity	Does the interpretation survive reasonable choices?
External prediction	Do topic shares predict an outcome outside estimation?

Mueller & Rauh (2018): predict next-year conflict from newspaper topics.

Reference: ESS course; Grimmer, Roberts & Stewart (2022); Mueller & Rauh (2018)

Checking a topic label

STM Topic 4 · provisional label: “International affairs”

Evidence	What to check
Probability words	council, community, agreed, united, european
FREX words	argentine, commonwealth, nations, agreed, meeting
Same Thatcher speech	European Council meeting in Strasbourg; T4 = 97.9%
Other speeches	Read high-share texts and possible counterexamples

Decision: retain, narrow, broaden or reject the label.

One speech supports a label; validate it on other texts.

Reference: Ying, Montgomery & Stewart (2022)

Reading perplexity

Question: how well does the model reconstruct the observed words?

$$\text{perplexity} = \exp\left(-\frac{\sum_{d,v} x_{dv} \log p_{dv}}{\sum_{d,v} x_{dv}}\right), \qquad p_{dv} = \sum_{k} \theta_{dk}\beta_{kv}$$

4 · Negate the average, then exponentiate

Read the statistic	Classroom interpretation
Lower perplexity is better	Same data, vocabulary and calculation.
LDA and STM	Same documents, vocabulary and token-weighted formula
Lab R04b–R04c	Estimation words are reused: **training / in-sample fit**

Training fit does not measure prediction of unseen words.

Reference: Blei, Ng & Jordan (2003); Wallach et al. (2009); Roberts, Stewart & Tingley (2019)

Held-out prediction and topic meaning

Evidence	How to read it
Training perplexity ↓	Better reconstruction of the estimation words
Held-out prediction	Hide evaluation words before fitting, then score them
STM workflow	`make.heldout()` → fit → `eval.heldout()`; compare K with `searchK()`
Topic meaning	Read texts; check stability and independent evidence

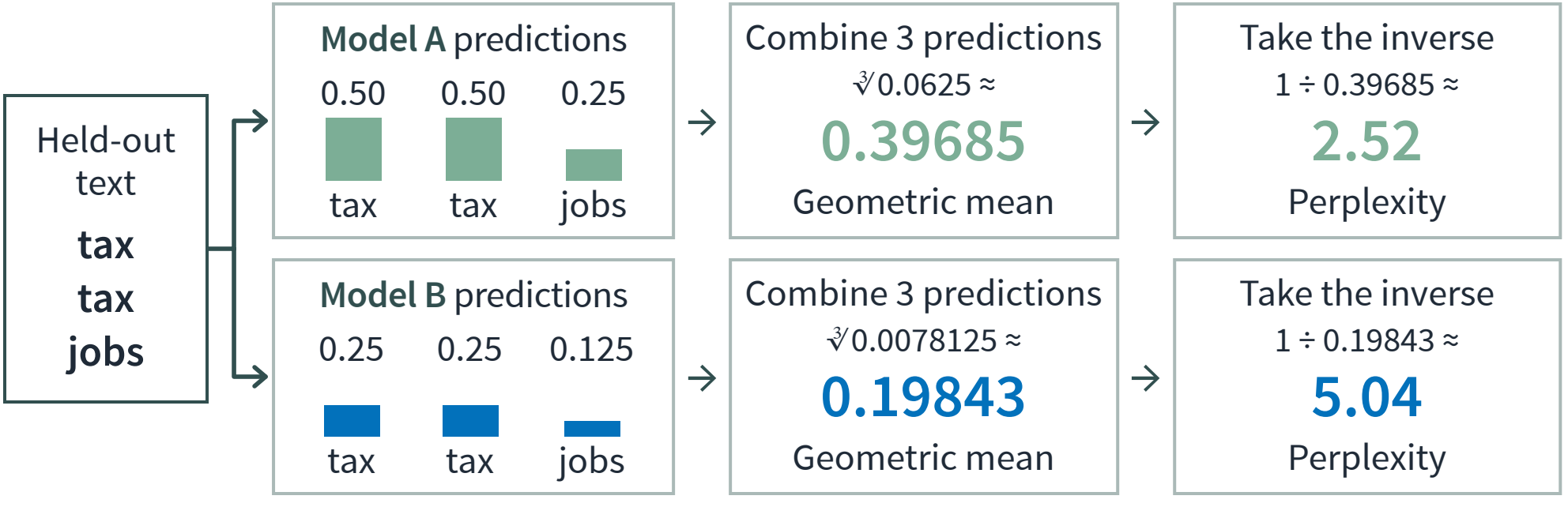
STM held-out log probability: higher is better. Perplexity: **lower is better.**

Chang et al. (2009): predictive fit and human interpretation can diverge.

Reference: Blei, Ng & Jordan (2003); Wallach et al. (2009); Roberts, Stewart & Tingley (2019); Chang et al. (2009)

Perplexity in a three-word example

Illustration: compare probabilities for the same hidden words.

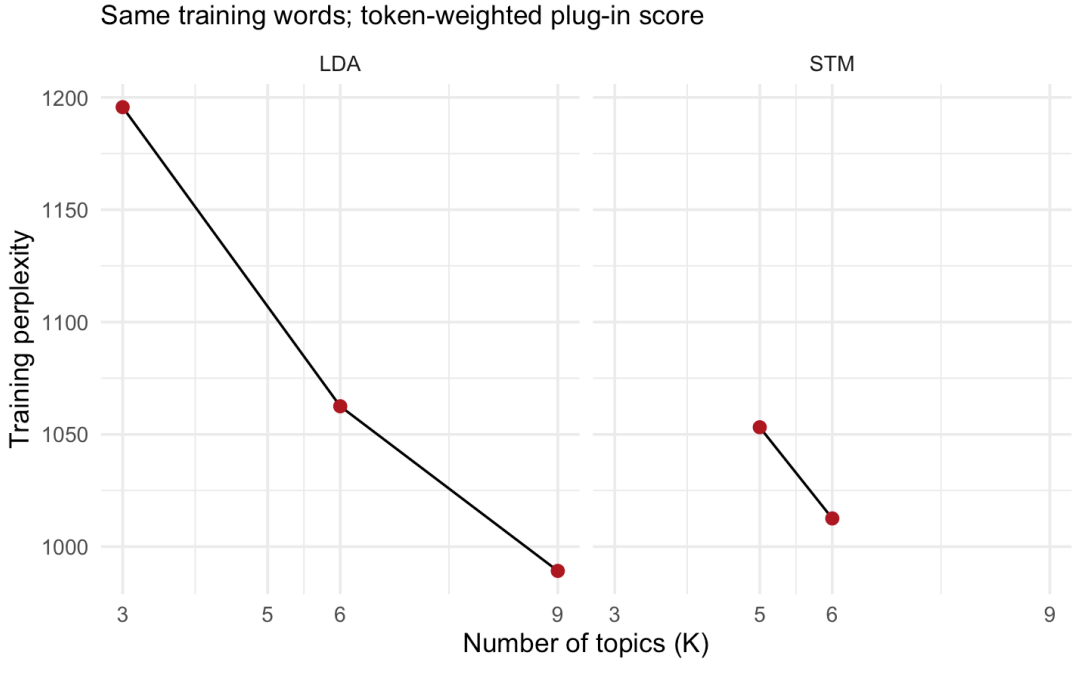


Higher word probabilities give lower perplexity.
Here, 2× the probability gives ½ the perplexity. Topic labels still need checking.

Reference: Blei, Ng & Jordan (2003); Wallach et al. (2009)

Checking the number of topics

Teacher demonstration · Choosing a topic resolution



- **LDA, K=3/6/9:** 1195.7 → 1062.5 → 989.2.
- **STM, K=5/6:** 1053.2 → 1012.6.
- Read words and the same texts: does extra K add a useful distinction?

Lower perplexity means better fit, not proof of the best K.

K=6 is our teaching resolution. R04b calculation · R04c plot.

Reference: Blei, Ng & Jordan (2003); Chang et al. (2009)

Keyword-assisted topic models

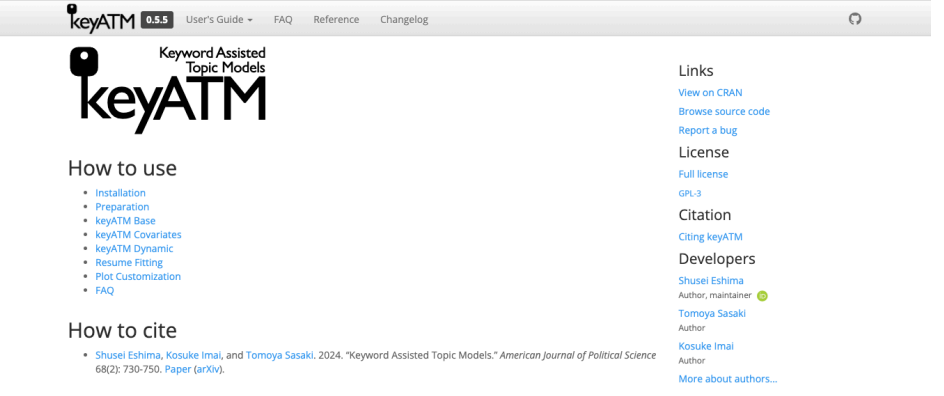
Another modelling choice · keyATM

Method paper



Eshima, Imai & Sasaki · AJPS 2024

R package




```
install.packages("keyATM")
```

Researchers supply a few keywords for concepts they want to measure. The model estimates topics and document mixtures, including topics without keywords.

Reference: Eshima, Imai & Sasaki (2024)

Extension · Keyword guidance in two models

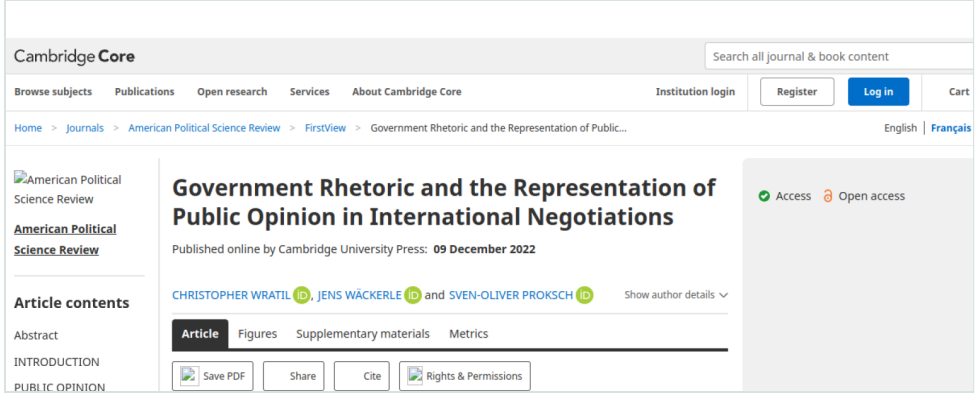
Method	Researcher input	W8 use
keyATM	Keywords for defined topics, plus topics without keywords	Method and package overview
Seeded LDA	Seed dictionary, pseudo-count weight and residual topics	Independent extension

Both use substantive information to guide topic discovery.

The models and outputs differ; interpret and validate each fit.

keyATM and seededlda. ESS Day 7's runnable example uses `textmodel_seededlda()`.

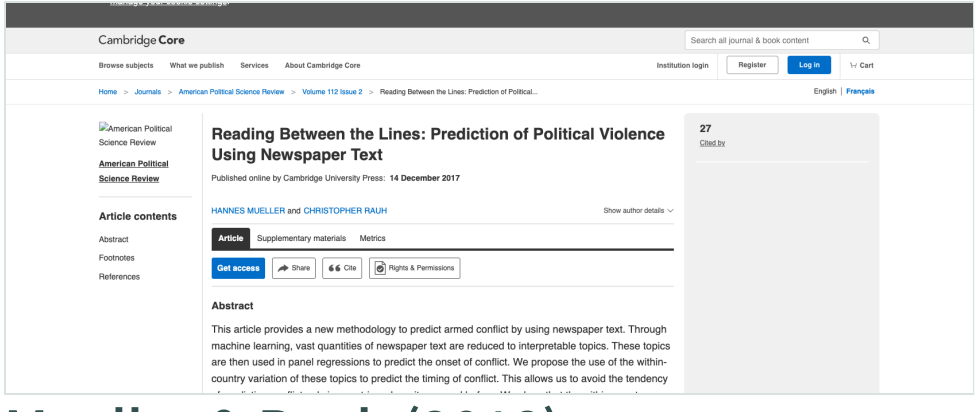
More applications in political research



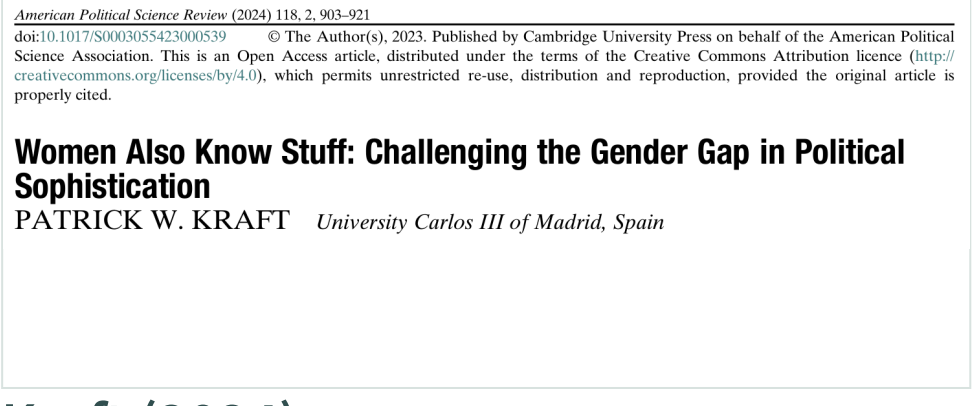
Wratil et al. (2023)



Motolinia (2021)



Mueller & Rauh (2018)



Kraft (2024)

Reference: Wratil et al. (2023); Motolinia (2021); Mueller & Rauh (2018); Kraft (2024)

Extension · Correlated topics and change over time

Model	Research question	What the analyst supplies
Correlated Topic Model	Which issues tend to appear together in documents?	Text counts and a chosen topic resolution
Dynamic Topic Model	How does the language of an issue change across periods?	Comparable texts and ordered time periods

CTM estimates which issues appear together. DTM tracks how topic vocabulary changes across ordered periods.

Reference: Blei & Lafferty (2007); Blei & Lafferty (2006)

Extension · Short texts with a biterm model

Short-text applications · R package BTM

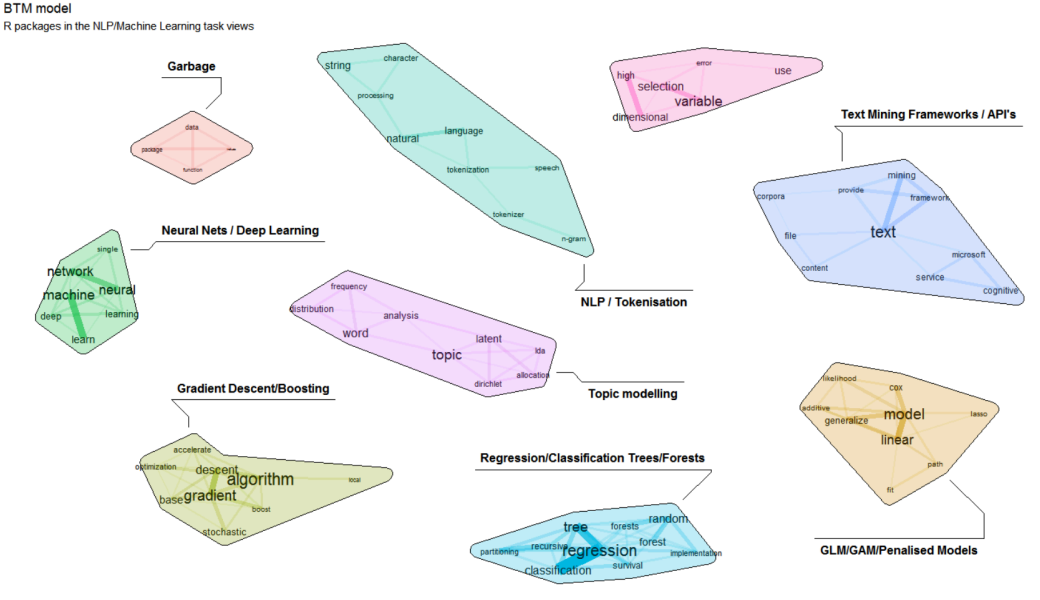
Original paper · WWW 2013

A Biterm Topic Model for Short Texts

Xiaohui Yan, Jiafeng Guo, Yanyan Lan, Xueqi Cheng
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- **Texts:** short survey answers, posts or headlines.
- **BTM:** pools unordered word pairs across short texts.

R package example



R package descriptions, grouped into topics.

Your extension: compare BTM with LDA and check labels against the original texts.

Reference: Yan et al. (2013); BTM documentation; official R package documentation

Break

Back at 17:36.

Next: the RStudio project, complete topic mixtures and party comparisons.

Student Lab ZIP includes the RStudio project, step-by-step R script, data and saved models.

One speech as one document

Corpus choice	Shared classroom input
Source	Commons leader speeches from two parties
Eligible source texts	3,897 nonempty speeches, source terms ≥ 80
Fixed teaching sample	719 complete speeches, up to 30 per party-period
Sample dates	15 May 1979 to 17 December 2025
Model input	Same document IDs across the fitted R models

Discuss: what changes if every party's speeches become one document?

Data source: Essex Summer School 2026, Day 7; British House of Commons party-leader speeches. W8 sampling seed 20261028. Nearly balanced party-period sample, not parliamentary word volume.

R workflow and model objects

Class stage	Steps	What you should explain
1 · Read and prepare	R01a–R01c	Define a document; explain what cleaning changes
2 · Fit and read LDA	R02a–R02b	Read a complete topic mixture
3 · Check a label	R03a–R03b; E02	Support a label with text and name a challenge
4 · Compare with STM	R05a–R05e; E05	Interpret direction, percentage points and uncertainty

Same texts: LDA uses counts; STM adds metadata.

Before class: R00 packages and project setup. Open `W8_lab.Rproj`, then `W8.R`.

After class: R03c–R04c compare party summaries and K diagnostics. The project plan and reading close the deck.

Student Lab · Student ZIP.

R00a: R packages

Before class · Run R00a in `W8.R`.

Code

R packages add functions for text analysis, models and plotting.

```
required <- c("quanteda", "quanteda.textstats",
"seededlda", "stm",
  "dplyr", "tidyr", "ggplot2", "jsonlite")
missing <- required[!vapply(required,
requireNamespace,
  quietly = TRUE, FUN.VALUE = logical(1))]
if (length(missing)) stop("Install these
packages first: ",
  paste(missing, collapse = ", "))
invisible(lapply(required, function(p)
  suppressPackageStartupMessages(library(p,
character.only = TRUE))))
options(width = 100, seededlda_threads = 1)
cat("Packages loaded:", paste(required, collapse
= ", "), "\n")
```

Output

```
Packages loaded: quanteda,
quanteda.textstats, seededlda, stm,
dplyr, tidyr, ggplot2, jsonlite
```

Before class · Check: If you see Install these packages first, install the listed packages and rerun this step.

R00b: Project folders

Before class · Run R00b in W8.R.

Code

```
lab_file <- function(name)
file.path("analysis", "ess_lab", name)
if
(!file.exists(lab_file("sample_raw.rds")
)) {
  stop("Open W8_lab.Rproj, then run R00b
again.")
}
party_colors <- c(Conservative =
"#003B73", Labour = "#B00020")
cat("W8 speech input found. Ready to
build and fit both models.\n")
```

Output

```
W8 speech input found. Ready to build and fit
both models.
```

Before class · Check: If the file check fails, reopen W8_lab.Rproj and rerun R00b.

R01a: Read file

Run together · Run **R01a** in `W8.R`.

Code

```
topic_docs <-
readRDS(lab_file("sample_raw.rds"))
topic_corpus <- corpus(
  topic_docs, text_field = "text",
docid_field = "doc_id")
print(topic_docs[1:3,
  c("doc_id", "leader", "party_family",
"date")])
print(c(speeches = ndoc(topic_corpus)))
```

Output

Document	Leader	Party	Date
ess_hoc_00002	Margaret Thatcher	Conservative	1979-05-15
ess_hoc_00211	James Callaghan	Labour	1979-06-21
ess_hoc_00004	Margaret Thatcher	Conservative	1979-06-26

Corpus: **719 complete speeches**.

Run together · Check: Confirm that the corpus contains 719 speeches and identify the document ID, leader, party, and date.

R step-by-step Lab R01a. Same 719 speeches. Full step and output. Reformatted first three records and the executed corpus count.

R01b: Split and clean words

Run together · Run **R01b** in `W8.R`.

Code

```
id <- "ess_hoc_00004"
raw_tokens <- tokens(topic_corpus[id], what =
"word")
parliamentary_stopwords <- c("hon", "member",
"right", "friend", "sir",
  "gentleman", "lady", "house", "commons",
"speaker", "minister",
  "secretary", "asked", "question", "mr", "mrs",
"ms", "â", "ã")
clean_tokens <- tokens(topic_corpus,
remove_punct = TRUE,
  remove_symbols = TRUE, remove_numbers = TRUE,
remove_url = TRUE,
  remove_separators = TRUE, split_hyphens =
FALSE) |>
  tokens_tolower() |>
tokens_remove(stopwords("en")) |>
  tokens_remove(parliamentary_stopwords)
print(raw_tokens[[id]][1:21])
print(clean_tokens[[id]][1:8])
```

Output

Raw tokens from `ess_hoc_00004`

```
"With" "permission" "," "Mr" "."
"Speaker"
"I" "should" "like" "to" "make" "a"
"statement" "on" "the" "European"
"Council" "Meeting"
"at" "Strasbourg" "."
```

After corpus cleaning

```
"permission" "like" "make" "statement"
"european" "council"
"meeting" "strasbourg"
```

Run together · Check: Explain the visible changes between the raw and cleaned words.

R step-by-step Lab R01b. Same 719 speeches. Full step and output. All executed raw and cleaned tokens from the same speech, reflowed six per line. Full Console formatting remains in the Lab.

R01c: Build the DFM

Run together · Run **R01c** in `W8.R`.

Code

```
collocations <- textstat_collocations(
  clean_tokens, min_count = 8, size = 2:3) |>
arrange(desc(lambda))
model_tokens <- tokens_compound(clean_tokens,
  phrase(head(collocations$collocation, 30)))
topic_dfm <- dfm(model_tokens) |>
  dfm_trim(min_termfreq = 5, min_docfreq = 2) |>
  dfm_trim(max_docfreq = .75, docfreq_type =
"prop")
topic_dfm <- topic_dfm[ntoken(topic_dfm) > 0, ]
topic_docs <-
topic_docs[match(docnames(topic_dfm),
topic_docs$doc_id), ]
topic_docs$retained_tokens <-
as.numeric(ntoken(topic_dfm))
stopifnot(identical(topic_docs$doc_id,
docnames(topic_dfm)))
print(c(speeches = ndoc(topic_dfm), features =
nfeat(topic_dfm),
  tokens = sum(ntoken(topic_dfm))))
print(topfeatures(topic_dfm, 6))
```

Output

```
speeches features   tokens
     719     2831    70813
     prime government     people
can          now    country
       992          728         590
490          382         354
```

Run together · Check: Confirm the three DFM totals. Explain how compounding and frequency trimming change what the models can see.

R step-by-step Lab R01c. Same 719 speeches. Full step and output. Executed DFM dimensions, retained-token total and all six most frequent retained features.

R02a: Fit six-topic LDA

Run together · Run **R02a** in `W8.R`.

Code

```
set.seed(20260710)
lda_6 <- textmodel_lda(topic_dfm, k = 6,
alpha = .5, beta = .1,
  max_iter = 500, auto_iter = FALSE,
batch_size = 1, verbose = FALSE)
print(terms(lda_6, 10))
print(dim(lda_6$theta))
```

Output

LDA topic	First five of ten printed words
T1	european, council, community, countries, agreed
T2	government, state, made, members, service
T3	prime, people, government, said, can
T4	government, people, unemployment, economy, year
T5	people, can, country, want, support
T6	united, must, international, security, nations

`theta` dimensions: **719 speeches × 6 topics**.

Run together · Check: Choose two topics and write provisional labels. Later, check those labels against actual speeches.

R step-by-step Lab R02a. Same 719 speeches. Full step and output. Selected first five words from each printed distribution. The Lab retains all ten.

R02b: Complete topic mixtures

Run together · Run **R02b** in `W8.R`.

Code

```
print(round(head(lda_6$theta, 3), 3))
print(range(rowSums(lda_6$theta)))
dominant_topic <- max.col(lda_6$theta,
ties.method = "first")
names(dominant_topic) <-
topic_docs$doc_id
print(head(dominant_topic, 5))
print(table(dominant_topic))
```

Output

Speech	T1	T2	T3	T4	T5	T6
00002	0.044	0.188	0.049	0.197	0.470	0.053
00211	0.005	0.490	0.136	0.207	0.116	0.045
00004	0.874	0.067	0.012	0.002	0.012	0.032

Dominant-topic counts (T1–T6): **91, 80, 193, 143, 145, 67**.

Full θ rows sum to **1**. Ties use the first index.

Run together · Check: Which output can answer: Which issues does one speech discuss as a mixture?

R step-by-step Lab R02b. Same 719 speeches. Full step and output. Reformatted first three θ rows and deterministic dominant-topic summary.

R03a: Two selected speeches

Run together · Run **R03a** in `W8.R`.

Code

```
theta <- lda_6$theta
high_i <- which.max(theta[, 1])
mixed_i <- which.min(apply(theta, 1,
max))
print(topic_docs[c(high_i, mixed_i),
  c("doc_id", "leader", "party_family",
"date")])
print(round(theta[c(high_i, mixed_i), ],
4))
print(rowSums(theta[c(high_i, mixed_i),
, drop = FALSE]))
```

Output

A: `ess_hoc_00004`, Thatcher, 26 June 1979.

B: `ess_hoc_05895`, Corbyn, 25 March 2020.

Speech	T1	T2	T3	T4	T5	T6
00004	0.8738	0.0668	0.0124	0.0025	0.0124	0.0322
05895	0.0982	0.1339	0.2054	0.2232	0.2054	0.1339

Both complete mixtures sum to **1** before display rounding.

Run together · Check: How does retaining only the largest topic change the description compared with keeping all six proportions?

R03b: Original speech texts

Run together · Run **R03b** in `W8.R`.

Code

```
cat("Document:",
topic_docs$doc_id[high_i], "\n")
cat(topic_docs$text[high_i], "\n")
cat("Mixed document:",
topic_docs$doc_id[mixed_i], "\n")
cat(topic_docs$text[mixed_i], "\n")
```

Output

A: `ess_hoc_00004`

> With permission, Mr. Speaker I should like to make a statement on the European Council Meeting at Strasbourg. With my…

B: `ess_hoc_05895`

> My hon. Friend understands the issue and represents his constituency extremely well. I know that a lot of his constituents…

Run together · Check: Find support and a possible challenge to the label. Complete E02; finish E03 after class.

R step-by-step Lab R03b. Same 719 speeches. Full step and output. Opening excerpts from the exact two printed source texts. Complete output in Lab.

R05a: Fit STM

Run together · Run **R05a** in `W8.R`.

Code

```
metadata <- data.frame(
  doc_id = topic_docs$doc_id,
  party_family = topic_docs$party_family,
  period = topic_docs$period)
design <- model.matrix(~ party_family + period,
metadata)
stopifnot(qr(design)$rank == ncol(design))
stm_input <- convert(topic_dfm, to = "stm")
stm_6 <- stm(stm_input$documents, vocab =
stm_input$vocab, K = 6,
  prevalence = ~ party_family + period, data =
metadata,
  seed = 20260710, init.type = "Spectral",
max.em.its = 100,
  emtol = 1e-4, verbose = FALSE)
print(labelTopics(stm_6, topics = 4, n = 5))
```

Output

STM topic 4. K=6, prevalence ~ `party_family + period`.

Word summary	Five printed words
Highest Prob	council, community, agreed, united, european
FREX	argentine, commonwealth, nations, agreed, meeting
Lift	dependencies, falklands, islanders, moore, shareholders
Score	names, argentine, council, islands, commonwealth

Run together · Check: Use the probability words and FREX words to assess the interpretation of the same STM topic.

R step-by-step Lab R05a. Same 719 speeches. Full step and output. All four labelTopics summaries printed by this step.

R05b: Raw STM party means

Run together · Run **R05b** in `W8.R`.

Code

```
stm_theta <- as.data.frame(stm_6$theta)
colnames(stm_theta) <- paste0("topic_",
seq_len(ncol(stm_theta)))
stm_theta$party_family <-
topic_docs$party_family
stm_party_means <- stm_theta |>
pivot_longer(starts_with("topic_"),
  names_to = "topic", values_to = "proportion")
|>
  group_by(party_family, topic) |>
  summarise(mean = mean(proportion), .groups =
"drop")
stm_raw_differences <- stm_party_means |>
  pivot_wider(id_cols = topic, names_from =
party_family, values_from = mean) |>
  mutate(difference = Labour - Conservative)
print(stm_raw_differences[stm_raw_differences$to
pic == "topic_4", ])
```

Output

```
# A tibble: 1 × 4
  topic   Conservative Labour
difference
  <chr>          <dbl>  <dbl>
<dbl>
1 topic_4        0.137 0.0763
-0.0603
```

Run together · Check: State what positive and negative values mean before moving to the conditional comparison.

R05c: Estimate party differences

Run together · Run **R05c** in `W8.R`.

Code

```
set.seed(20260710)
stm_party_effects <- estimateEffect(1:6 ~
party_family + period,
  stmobj = stm_6, metadata = metadata,
  uncertainty = "Global", nsims = 50)
set.seed(20260710)
adjusted_result <- plot(stm_party_effects,
  covariate = "party_family", method =
"difference",
  cov.value1 = "Labour", cov.value2 =
"Conservative",
  topics = 1:6, nsims = 1000, omit.plot = TRUE,
  labeltype = "numbers")
print(adjusted_result$means[4])
print(adjusted_result$cis[[4]])
```

Output

```
[[1]]
[1] -0.05993053

        2.5%        97.5%
-0.09272104 -0.02778845
```

Run together · Check: Which layer of uncertainty does this interval describe? Does it cover every vocabulary, sample, or naming choice?

R05d: Comparison table

Run together · Run **R05d** in `W8.R`.

Code

```
adjusted_differences <- data.frame(topic
= paste0("topic_", 1:6),
  difference =
unlist(adjusted_result$means),
  lower = vapply(adjusted_result$cis,
`[`, numeric(1), 1),
  upper = vapply(adjusted_result$cis,
`[`, numeric(1), 2))
print(adjusted_differences[adjusted_diff
erences$topic == "topic_4", ],
  row.names = FALSE)
```

Output

```
topic  difference       lower       upper
 topic_4 -0.05993053 -0.09272104 -0.02778845
```

Run together · Check: Check the topic 4 values before plotting them in R05e. Complete E05 after R05e by separating a supported association from a causal claim the data cannot identify.

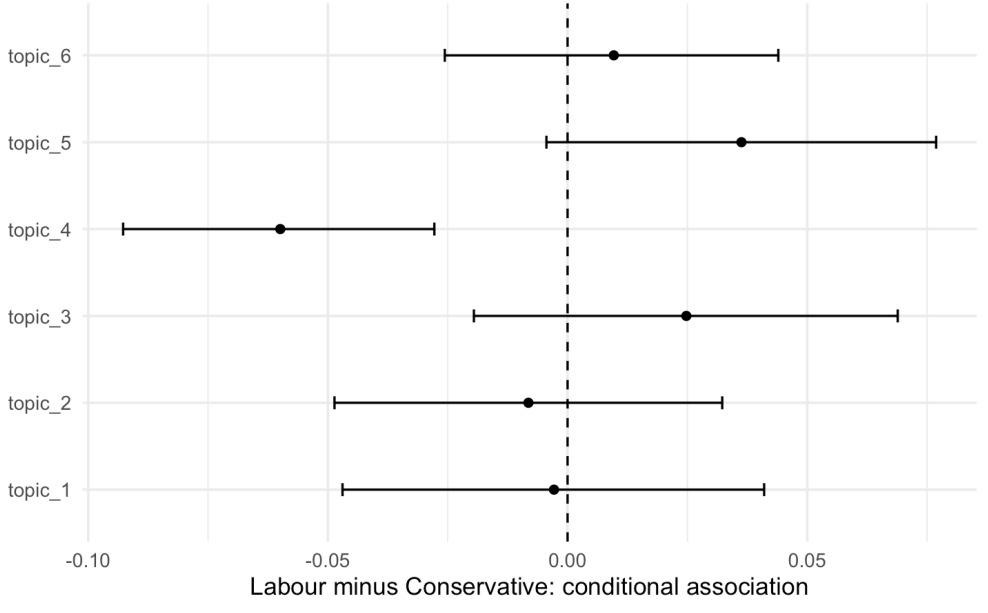
R05e: STM interval plot

Run together · Run **R05e** in `W8.R`.

Code

```
p_stm_adjusted <-
ggplot(adjusted_differences,
aes(difference, topic)) +
  geom_vline(xintercept = 0, linetype =
2) + geom_point() +
  geom_errorbar(aes(xmin = lower, xmax =
upper),
    orientation = "y", width = .15) +
  labs(x = "Labour minus Conservative:
conditional association", y = NULL) +
  theme_minimal(base_size = 16)
print(p_stm_adjusted)
```

Output



Run together · **Check:** Intervals refer to the same fitted STM.

R step-by-step Lab R05e. Same 719 speeches. Full step and output. Research design. Actual R plot from the exact displayed step. Console values and the full plot are in the Lab.

R03c: LDA party means

After class · Run **R03c** in `W8.R`.

Code

```
lda_theta <- as.data.frame(theta)
colnames(lda_theta) <- paste0("topic_",
seq_len(ncol(theta)))
lda_theta$doc_id <- topic_docs$doc_id
lda_theta$party_family <-
topic_docs$party_family
lda_theta_long <- lda_theta |>
pivot_longer(starts_with("topic_"),
  names_to = "topic", values_to =
"proportion")
party_topic_means <- lda_theta_long |>
group_by(party_family, topic) |>
  summarise(mean_proportion =
mean(proportion), .groups = "drop")
print(as.data.frame(party_topic_means),
row.names = FALSE)
```

Output

LDA topic	Conservative	Labour
T1	0.1510	0.1174
T2	0.1311	0.1532
T3	0.2245	0.2323
T4	0.1745	0.1870
T5	0.2171	0.2034
T6	0.1019	0.1067

After class · Check: What quantity would you estimate if speeches were weighted by retained_tokens?

R step-by-step Lab R03c. Same 719 speeches. Full step and output. Same twelve printed means, reshaped to a wide table and rounded to four decimals.

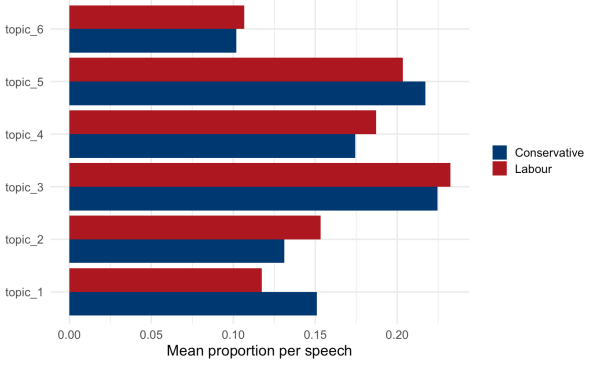
R03d: LDA party plot

After class · Run R03d in W8.R.

Code

```
p_lda_party <- ggplot(party_topic_means,
  aes(topic, mean_proportion, fill =
party_family)) +
  geom_col(position = "dodge") +
coord_flip() +
  scale_fill_manual(values =
party_colors) +
  labs(x = NULL, y = "Mean proportion
per speech", fill = NULL) +
  theme_minimal(base_size = 16)
print(p_lda_party)
```

Output



- **Topic 1:** Conservative 15.1%; Labour 11.7%.
- Each sampled speech has equal weight.
- These means are unadjusted for period and do not identify a causal effect.

R step-by-step Lab R03d. Same 719 speeches. Full step and output. Actual R03d plot; the Topic 1 percentages are the same party means calculated in R03c.

R04a: Alternative resolutions

After class · Run R04a in W8.R.

Code

```
k_candidates <- c(3, 6, 9)
lda_models <- lapply(k_candidates,
function(k_value)
  readRDS(lab_file(paste0("lda",
k_value, ".rds"))))
names(lda_models) <- paste0("K",
k_candidates)
stm_models <- lapply(c(5, 6),
function(k_value)
  readRDS(lab_file(paste0("stm",
k_value, ".rds"))))
print(lapply(lda_models, terms, 5))
```

Output

LDA topic	Five printed words
K=3, T1	european, council, community, countries, united
K=6, T1	european, council, community, countries, agreed
K=9, T3	european, union, deal, europe, future
K=9, T5	council, agreed, meeting, community, european

Check: Match words before comparing topic numbers.

R step-by-step Lab R04a. Same 719 speeches. Full step and output. Four selected columns from the printed K=3/6/9 word lists. All columns remain in Lab.

R04b: Training perplexity

After class · Run **R04b** in `W8.R`.

Code

```
training_perplexity <- function(model, data) {
  is_stm <- inherits(model, "STM")
  vocab <- if (is_stm) model$vocab else
colnames(model$phi)
  phi <- if (is_stm)
exp(model$beta$logbeta[[1]]) else model$phi
  counts <- as.matrix(data[, vocab])
  probabilities <- pmax(model$theta %*% phi, 1e-
12)
  exp(-sum(counts * log(probabilities)) /
sum(counts))
}
perplexity_results <- data.frame(Model =
c("LDA", "LDA", "LDA", "STM", "STM"),
  k = c(3, 6, 9, 5, 6), perplexity =
vapply(c(lda_models, stm_models),
    training_perplexity, numeric(1), data =
topic_dfm))
print(perplexity_results, row.names = FALSE)
```

Output

Model	K	Training perplexity
LDA	3	1195.66
LDA	6	1062.50
LDA	9	989.21
STM	5	1053.16
STM	6	1012.57

Lower: better reconstruction of these training words.

Same DFM and token-weighted formula; check topic meaning separately.

R step-by-step Lab R04b. Same 719 speeches. Full step and output. All five executed scores, rounded to two decimals; full precision and the common formula remain in the Lab. Training fit does not establish a superior model family or the best K.

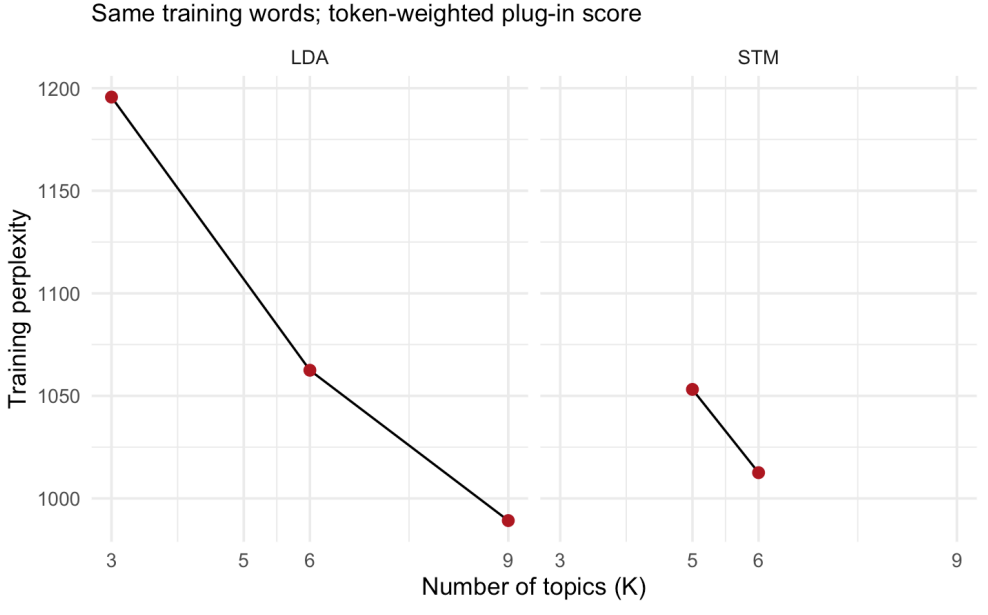
R04c: Training-fit plot

After class · Run **R04c** in `W8.R`.

Code

```
p_perplexity <-
ggplot(perplexity_results, aes(k,
perplexity)) +
  geom_line() + geom_point(color =
"#B00020", size = 3) +
  facet_wrap(~ Model) +
scale_x_continuous(breaks = c(3, 5, 6,
9)) +
  labs(x = "Number of topics (K)", y =
"Training perplexity",
    subtitle = "Same training words;
token-weighted plug-in score") +
  theme_minimal(base_size = 16)
print(p_perplexity)
```

Output



After class · Check: Fit uses the same training DFM.

R step-by-step Lab R04c. Same 719 speeches. Full step and output. Actual R plot from the exact displayed step. Console values and the full plot are in the Lab.

Extended comparison: Issue classification

Research step	How this study proceeds
Specify labels	Define the issue categories before fitting
Train and validate	Use separately labelled news and tweet samples
Measure attention	Aggregate classified texts to daily group shares
Compare timing	Model how groups' earlier attention predicts later attention
Interpret	Check differences across actors and issues

Compare: Barberá et al. discover themes; this study starts with issue labels.

Reference: Gilardi, Gessler, Kubli & Müller (2022); Barberá et al. (2019)

A validation plan for your project

Your research design · four minutes

- Define the concept, population and one document.
- Choose LDA or STM; explain what information you supply.
- State the comparison, denominator and weighting.
- Specify a sensitivity check and independent evidence.

Share one failure case: what evidence would make you change the topic label or abandon the measure?

Reference: Grimmer, Roberts & Stewart (2022)

Reading and the next class

Core: *Text as Data*, ch. 13; selected §§21.3–21.4.

Further reading	Research use
Wratil et al. (2023)	Government rhetoric and public opinion
Mueller & Rauh (2018)	Next-year conflict prediction
Roberts et al. (2014)	Survey answers and human coding
Eshima et al. (2024)	Keyword-assisted measurement
Gilardi et al. (2022)	Supervised issue classification

Practice: E01–E05. Optional diagnostics appear before these closing pages.

Next: readability, similarity and word embeddings.

References: Methods and validation

Catalinac, A. 2016. “From Pork to Policy: The Rise of Programmatic Campaigning in Japanese Elections.” *JOP* 78(1): 1–18. DOI.

Blei, D. M., A. Y. Ng, and M. I. Jordan. 2003. “Latent Dirichlet Allocation.” *JMLR* 3: 993–1022. Article.

Roberts, M. E., et al. 2014. “Structural Topic Models for Open-Ended Survey Responses.” *AJPS* 58(4): 1064–1082. DOI.

Eshima, S., K. Imai, and T. Sasaki. 2024. “Keyword-Assisted Topic Models.” *AJPS* 68(2): 730–750. DOI.

Original-study figures retain the source versions and page locators printed on their slides.

References: Readings and reproducibility

Mueller, H., and C. Rauh. 2018. “Reading Between the Lines: Prediction of Political Violence Using Newspaper Text.” *APSR* 112(2): 358–375. DOI.

Ying, L., J. M. Montgomery, and B. M. Stewart. 2022. “Topics, Concepts, and Measurement.” *Political Analysis* 30(4): 570–589. DOI.

Grimmer, J., M. E. Roberts, and B. M. Stewart. 2022. *Text as Data*. Princeton University Press. Ch. 13; selected §§21.3–21.4. Contents.

Schoonvelde, M. 2026. *Introduction to Quantitative Text Analysis*. Essex Summer School, Day 7 slides and lab. ESS course.

W8 classroom practical: LDA and STM from texts to findings. Full Lab · Student lab ZIP.

W8 sources, model recipes and input metadata accompany the classroom materials. Presenter notes remain outside the student package.

References: Variants and model evaluation

Blei, D. M., and J. D. Lafferty. 2007. “A Correlated Topic Model of Science.” *Annals of Applied Statistics* 1(1): 17–35. DOI.

Blei, D. M., and J. D. Lafferty. 2006. “Dynamic Topic Models.” *ICML*: 113–120. DOI.

Chang, J., et al. 2009. “Reading Tea Leaves: How Humans Interpret Topic Models.” *NeurIPS 22*. Paper.

Wallach, H. M., et al. 2009. “Evaluation Methods for Topic Models.” *ICML*: 1105–1112. DOI.

Roberts, M. E., B. M. Stewart, and D. Tingley. 2019. “stm: An R Package for Structural Topic Models.” *JSS* 91(2): 1–40. DOI.

Original methods and maintained software documentation. Method choices and output definitions are separate from political identification.

References: Applications and developments

Gilardi, F., C. R. Shipan, and B. Wüest. 2021. “Policy Diffusion: The Issue-Definition Stage.” *AJPS* 65(1): 21–35. DOI.

Wratil, C., J. Wäckerle, and S.-O. Proksch. 2023. “Government Rhetoric and the Representation of Public Opinion in International Negotiations.” *APSR* 117(3): 1105–1122. DOI.

Barberá, P., et al. 2019. “Who Leads? Who Follows?” *APSR* 113(4): 883–901. DOI.

Bernhard-Harrer, J., et al. 2026. “Beyond Standardization: A Comprehensive Review of Topic Modeling Validation Methods for Computational Social Science Research.” *PSRM* 14(3): 833–851. DOI.

Issue publication years are shown. Follow the linked original articles for sources.